
THE GEOMETRIC NATURE OF LIGHT

*From Ancient Substrate to Quantum Field. A
Continuous-Field Reading of Twenty-Five Centuries of
Optical Inquiry*

Theoretical Physics · Philosophy of Science · Manuscript for External Review

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ABSTRACT

This paper traces a structural intuition across twenty-five centuries of optical inquiry: light is a localized perturbation propagating through a continuous medium that is space itself. The thread runs from the visual rays of Empedocles, through Aristotle's actualization of the transparent, the medieval Latin and Islamic optical traditions, the Newton-Huygens bifurcation, Maxwell's electromagnetic synthesis, and the twentieth-century revolution culminating in quantum field theory. The interpretive position the paper develops is the neo-Lorentzian reading of relativistic and quantum field-theoretic physics. The substrate is taken to be physically real and continuous. Lorentz-invariance is treated as an emergent feature of the substrate's dynamics rather than as a metaphysical primitive. Empirical predictions are identical to those of standard special relativity in tested regimes; ontological commitments are stronger and clearer.

The reading dissolves several interpretive paradoxes that the standard reading leaves open. Wave-particle duality is reframed as the meeting of a continuous substrate with discrete matter at the point of registration. Time dilation is reframed as a path-dependent rate of internal-process accumulation between shared boundary events, without invoking time as a substance or dimension. Cosmological metric expansion is understood as the substrate's bodily extension, distinct from and not bounded by the propagation rate of perturbations within it. Black-hole event horizons are global features of substrate geometry rather than local field-strength extrema. Closed-timelike-curve paradoxes dissolve at their premise on substrate-thermodynamic grounds.

The framework is empirically equivalent to standard physics in essentially all currently tested regimes. Section 11 enumerates eleven specific falsifiable predictions that distinguish the framework from standard interpretations in regimes where current observations are not yet definitive. Eight of these carry substrate-specific content; three are consistency tests with standard physics. Among the distinguishing predictions, the prohibition of phantom dark energy ($w(z) \geq -1$ strictly) is in active empirical tension with DESI DR1 and DR2 observations at the $2.5\text{-}3.9\sigma$ level, with the framework's prediction at risk of falsification within the next several years rather than at the formal Stage-IV timeline. The framework is not protected from refutation by its interpretive character; the interpretive commitments produce predictive consequences in regimes where current physics is not definitive, and those consequences are testable.

Keywords: light, electromagnetic field, quantum field theory, neo-Lorentzian relativity, substrate ontology, continuous field, photon, wave-particle duality, dark energy, MOND, Hubble tension, cosmology, philosophy of physics

1. INTRODUCTION: WHY LIGHT IS THE QUESTION OF SUBSTRATE

Every serious theory of light is, beneath its surface, a theory of what space is. From the visual rays of Empedocles to the photon of Einstein, the central problem has always been the same: when something is seen, what travels, and through what does it travel? The history of optics is the history of answers to those two questions, and the history of those answers is the history of how successive cultures have understood the substance of space itself.

This paper argues that the long sequence of optical theories, from antiquity through the present, converges on a structural intuition that has been articulated under increasingly precise mathematical vocabularies: light is a localized perturbation propagating through a continuous medium that is space itself. The medium is not a thing within space. The medium is space, considered as a substrate that admits perturbations. The propagation of those perturbations occurs at a finite rate set by the substrate's intrinsic properties. The medium also has its own bodily extension, which can proceed at rates entirely independent of, and frequently exceeding, the propagation rate of perturbations within it.

The structural intuition is continuous across the historical sequence; the specific theories embedding it are not. Aristotle's instantaneous activation of the transparent and Maxwell's finite-speed electromagnetic wave are different theories, not the same theory at different levels of formal precision. The unifying thread is the commitment to a continuous medium and the rejection of the void as the locus of optical phenomena. The differences are real and the paper marks them where they matter.

The interpretive position the paper develops is closer to the Lorentz-Poincaré ether tradition than to the standard reading of special relativity. Both make identical empirical predictions in tested regimes; they differ in ontology. The Lorentzian flavor takes the substrate to be physically real and the apparent absence of a preferred frame to be a consequence of the substrate's dynamics making no preferred frame kinematically distinguishable, rather than a metaphysical claim that no such frame exists. The Einsteinian standard reading takes Lorentz-invariance as primitive and treats the absence of a preferred frame as fundamental. The choice between these is empirically underdetermined and philosophically substantive. This paper takes the Lorentzian interpretive option openly, acknowledges its empirical equivalence with standard SR, and proceeds.

This reading aligns naturally with quantum field theory taken as an ontology rather than merely as a calculational scheme. The photon in modern field theory is an excitation of the electromagnetic field. The field is the substrate. The

excitation is the perturbation. The geometric content the paper articulates is what QFT's mathematics already encodes, expressed in language that emphasizes the substrate-perturbation structure rather than the operator-algebraic formalism. The contribution of the present paper is not a new prediction but a clearer articulation of the geometry that already underlies the most successful physics we have, an honest acknowledgment of the interpretive commitments this entails, and a set of falsifiable predictions that follow from the framework's specific ontological choices.

The paper proceeds historically. Section 2 treats the ancient conception, from Empedocles through the Hellenistic and Islamic optical traditions. Section 3 follows the medieval and early modern refinements through the Newton-Huygens bifurcation. Section 4 traces the wave theory's nineteenth-century triumph and the ether crisis. Section 5 covers the twentieth-century revolution, from Einstein's photon to quantum field theory. Section 6 states the substrate-perturbation reading explicitly. Section 7 corrects a frequently encountered error about the relation between propagation speed and frequency. Section 8 treats the modern phenomena that the reading must accommodate. Section 9 recovers the Aristotelian definition of time and shows how it dissolves the time-travel paradoxes. Section 10 notes open empirical questions. Section 11 presents specific falsifiable predictions and a summary table of testable claims. Section 12 concludes.

For readers familiar with the framework from which this paper draws (the system of working hypotheses the author terms "Trisduction"), a small number of internal terms appear in parentheses for cross-reference. Readers unfamiliar with that vocabulary may safely ignore the parentheticals; the geometric argument is complete in plain language.

2. THE ANCIENT CONCEPTION OF LIGHT

2.1 The Pre-Socratic Origins

The earliest Greek theories of vision and light were extramissive. Empedocles of Acragas (fifth century BCE) proposed that the eye projected a fire outward, mingling with a fire emitted by visible bodies, and that vision occurred at the meeting point of the two streams. The doctrine survives in fragments preserved by Aristotle and Plutarch. Plato echoed it in the *Timaeus*, where the gods are said to have wrought the eye such that "the pure fire which is within us, akin to that of day, flows through the eyes" and meets the external light "to make one body in the direction of the eyes."

Empedoclean ray-theory was wrong about direction (vision is intromissive, not extramissive) but right about something deeper: it located vision in a continuous interaction between observer and observed, mediated by something that filled the intervening space. The "fire" that streamed from the eye and the "fire" that streamed from the object met in a continuous medium. This is the seed of every subsequent field theory of light, mistaken in its details but correct in its commitment to a continuous mediator.

The atomistic alternative, articulated most fully by Leucippus and Democritus, treated vision as the reception of thin material films (*eidola*) sloughed off by visible bodies. Lucretius (first century BCE) gave the atomistic theory its mature literary expression in *De Rerum Natura*, Book IV. The simulacra travel through the void, strike the eye, and produce vision by direct mechanical contact. The atomistic theory was wrong about the mechanism but anticipated something that would matter much later: the discrete, packet-like aspect that vision and light would prove to have under measurement.

The two ancient traditions, ray-theory and simulacrum-theory, foreshadow the two faces of the modern photon. Light has both a continuous propagation in a medium and a discrete arrival at a detector. Antiquity divided what would later prove to be unified.

2.2 Aristotle and the Actualization of the Transparent

Aristotle, in *De Anima* Book II Chapter 7 and *De Sensu* Chapter 2-3, broke from both Empedoclean ray-theory and atomistic emission. He defined light not as a body emitted from anything, but as the *actualization of the transparent*. The transparent (τὸ διαφανές) is a property that water, air, and aether share. In its potency, the transparent is dark. When something colored or self-luminous activates it, it becomes light.

Aristotle's view has been misunderstood for centuries because of its terminology, but its geometry is precise. Light, for Aristotle, is not a thing that travels through a medium. Light is a state of the medium. The transparent, when activated, is light, just as water, when set in motion, is wave. The wave is not a separate object riding on the water. The wave is the water in motion. By the same token, light is not an object passing through air or aether. Light is the air or aether in its activated state.

This is a continuous-medium ontology in pre-formal language. Aristotle had the medium-substrate intuition (the transparent is continuous; light is a state of the medium) but his actual theory differs substantively from a finite- c propagation theory. He believed light propagated instantaneously, a position not refuted in his own terms because his theory predicted no measurable transit time. The transparent does not "carry" light from source to receiver; it becomes light wherever the activating condition holds. Modern measurement reveals a finite propagation rate, and adding finite c to Aristotle's framework changes the theory rather than completing it. An instantaneous-activation theory and a finite- c propagation theory are different theories, even when both posit a continuous medium. What the present paper retains from Aristotle is the medium-commitment and the rejection of the void as the locus of optical phenomena. What it does not retain is the instantaneity. The unifying thread is real but partial.

Aristotle's framework also contained an idea that recurs at every later stage of optical theory: the prohibition of the void. The transparent is continuous because, for Aristotle, all of nature is continuous. Where there is no medium, there is no light, because there is no transparent to be activated. This commitment, much abused in subsequent commentary, contains a truth that nineteenth- and twentieth-century physics would force back into prominence after centuries of neglect: light requires a medium, even if the medium is space itself.

2.3 Euclid and the Geometric Tradition

Euclid's *Optics*, written probably in the early third century BCE, is the first surviving treatise on geometric optics. It treats vision as a cone of rays issuing from the eye, with each visible point determined by a ray meeting it. The ray-tracing geometry is correct under reversal: whether the rays issue from the eye or arrive at the eye, the geometric relationships hold. Euclid's *Optics* introduced the rigorous treatment of perspective, apparent size, and shadow that would underlie all subsequent geometric optics.

Euclid's silence on the physical nature of light is not an evasion. It is a methodological choice. Geometric optics is independent of the physical theory of light's nature. Lines connect points whether the lines are streams of fire, films of atoms, or activations of a continuous medium. Euclid's heirs would inherit a

powerful geometric tool that could be combined with any underlying physics. This separation of the geometric and the physical has remained a strength of optical theory ever since.

2.4 Ptolemy and the Hellenistic Refinement

Claudius Ptolemy's *Optics* (second century CE), preserved in Latin translation from a lost Arabic version of the Greek original, advanced beyond Euclid in two important respects. Ptolemy treated refraction systematically, recording the first known tables of angles of incidence and refraction at air-water and air-glass interfaces. He also addressed binocular vision, color, and the perception of motion.

Ptolemy's refraction tables are imperfect (they imply a linear relationship between angles of incidence and refraction, which is false at large angles), but they represent the first sustained attempt to bring optics under quantitative empirical discipline. The discrepancy between Ptolemy's tables and the true law of refraction would not be resolved until the seventeenth century, when Snell and Descartes independently identified the sine law. But the project of measuring the bending of light at a boundary, and of producing tables that could be checked, marks the beginning of optics as an experimental science.

2.5 Alhacen and the Islamic Optical Revolution

The most consequential figure in pre-modern optics is Ibn al-Haytham, known in Latin as Alhacen, who worked in Cairo in the late tenth and early eleventh centuries. His *Kitab al-Manazir* (*Book of Optics*), composed between approximately 1011 and 1021, represents the most thorough single advance in the science of light before Newton. The treatise was translated into Latin in the late twelfth century and dominated European optical thought for the following four centuries.

Alhacen accomplished three things of fundamental importance. First, he overturned the extramissive theory definitively. Through arguments from afterimages, dazzling, and the analysis of image formation, he established that light arrives at the eye from external sources, not the reverse. The intromissive theory has held ever since.

Second, Alhacen instituted the experimental method in optics. The *Book of Optics* is structured around what we would now call controlled experiments. Alhacen described in precise detail his use of the camera obscura (he was the first to give it systematic theoretical treatment, though the principle was known earlier), his experiments on refraction with carefully prepared apparatus, his observations of solar and lunar eclipses through pinhole projection, and his investigations of color and afterimages. The methodology he established (state the question, design the experiment, record the observations, draw the

inference, check against further experiment) is the methodology of empirical science.

Third, Alhacen articulated a theory of vision that integrated geometric optics with a mechanical account of the eye and a psychological account of perception. Light enters the eye, refracts at the cornea and lens, and produces an image. The image is then processed by the visual system to yield perception. This three-stage account (transmission, optical formation, perceptual interpretation) is essentially the framework still used today.

Alhacen's significance for the present argument is that he restored to optics what Aristotle had grasped intuitively and the atomists had abandoned: the necessity of a continuous medium between source and observer. Alhacen's light travels in straight lines through air, refracts at boundaries, and arrives at the eye through a continuous spatial process. He did not formalize the medium as a substrate in the modern sense, but his geometric and physical accounts are consistent only on the assumption that there is something continuous through which light moves. The medieval Latin tradition that inherited his work would carry this commitment forward.

3. MEDIEVAL AND EARLY MODERN REFINEMENT

3.1 The Latin Optical Tradition

The Latin reception of Alhacen, mediated through the *Liber de Aspectibus* and through the digests of Witelo and Pecham, produced the perspectivist tradition that dominated late medieval optics. Roger Bacon's *Opus Majus* (1267), Witelo's *Perspectiva* (circa 1270), and John Pecham's *Perspectiva Communis* (circa 1280) systematized Alhacen's experimental geometry and integrated it with Aristotelian natural philosophy and Christian theology. Bacon, in particular, urged the optical sciences as the foundation of all natural knowledge, arguing that "without the science of perspective, nothing in the things of this world can be known."

Medieval optics was geometrically sophisticated and experimentally grounded but conservative on the underlying physics. The dominant view, descended from Aristotle through Alhacen, treated light as a quality of the transparent medium, propagating through space without itself being a body. The propagation was usually held to be instantaneous or nearly so, though some authors (notably Witelo) speculated that light might require a finite, if very short, time to traverse great distances.

The medieval contribution to optics was the consolidation and transmission of a coherent body of experimental knowledge that survived into the early modern period. When Kepler and Descartes returned to optical questions in the seventeenth century, they did so on a foundation built primarily by Alhacen and his Latin successors, not by classical antiquity directly.

3.2 Kepler and the Retinal Image

Johannes Kepler's *Ad Vitellionem Paralipomena* (1604) and *Dioptrice* (1611) brought a decisive precision to the geometric optics of vision. Kepler established that the image is formed on the retina, that it is inverted relative to the object, and that the lens of the eye performs a specific refractive function whose geometry he calculated. He also gave the first essentially correct account of how spectacles correct vision, and he extended the analysis to the telescope, which Galileo had recently put to astronomical use.

Kepler's optical work is geometric. He did not commit himself strongly to any particular theory of light's underlying nature, treating it as a "species" propagating in lines through a continuous medium. His silence on the physical question is consistent with his methodological caution: where calculation suffices, do not multiply ontology. The geometric structure of refraction and

image formation is independent of the physical theory of light, just as Euclid had implicitly recognized.

3.3 Descartes and the Mechanical Theory

René Descartes, in *La Dioptrique* (1637), proposed a mechanical theory of light. Light, for Descartes, was a pressure or impulse transmitted instantaneously through a plenum of subtle matter. The medium was continuous (Descartes shared Aristotle's prohibition of the void), and light was a tendency to motion communicated through this medium without any actual bodily motion of the medium itself.

Descartes's theory has been criticized for its instantaneity. He held that light's transmission is literally simultaneous, a position he defended on metaphysical grounds. He was wrong empirically; light has a finite speed. But his geometric formulations were powerful. Descartes derived the sine law of refraction (Snell having reached it independently a few years earlier) by treating refraction as analogous to the bending of a ball passing from one medium to another. The mechanical analogy was loose, but the resulting law is correct.

Descartes's importance for the present argument is twofold. First, he firmly established the mechanical-medium picture in the seventeenth century, making it the default framework against which Newton would react and Huygens would refine. Second, he articulated the demand that physics give a continuous mechanism for every observed effect, which Newton would deliberately resist and Huygens would pursue. The two responses to the Cartesian demand define the seventeenth-century debate.

3.4 Galileo's Failed Speed Measurement

Galileo, in the *Discorsi* (1638), reported an experiment he had attempted with assistants on distant hilltops, each holding a covered lantern. The plan was to uncover the lanterns in sequence and measure the time delay. Galileo found no measurable delay at the available distances and concluded only that light's speed, if finite, must be very great. He did not conclude that it was infinite, and he understood the limitation of his experiment.

The experiment is interesting for its method. Galileo treated light's speed as an empirical question to be settled by measurement, not a metaphysical issue to be settled by definition. He could not measure the speed because his apparatus and distances were inadequate. The first successful measurement of light's speed would come from Ole Rømer in 1676, using observations of Jupiter's moon Io. Rømer's value (about 220,000 kilometers per second, low by a factor of about a third) established that light is finite and that the question of its speed is empirical.

The significance of the Rømer measurement for the present argument is that it confirmed what Aristotle had not foreseen but what his framework can accommodate. The transparent has an intrinsic relaxation rate, and that rate is finite. The substrate has its own thermodynamics, and the propagation of activation through the substrate occurs at a definite, measurable speed.

4. THE NEWTON-HUYGENS BIFURCATION AND ITS RESOLUTION

4.1 Newton's Corpuscular Theory

Isaac Newton's *Opticks* (1704) presented the most comprehensive and experimentally rigorous treatment of light to date. Newton's experimental work on the prismatic decomposition of white light into colors, on the inflexion (diffraction) of light around obstacles, on the colors of thin films, and on the production and interpretation of spectra remains foundational. The *Opticks* is perhaps the greatest single experimental achievement in pre-twentieth-century physics.

Newton's underlying theory of light was corpuscular. He held that light consists of small particles emitted by luminous bodies, traveling in straight lines through space, and interacting with matter according to mechanical principles. Newton explained refraction by supposing that the medium attracted the corpuscles, increasing their speed at the boundary and bending their trajectory. He explained the production of colors by supposing that white light contains corpuscles of different sizes or vibratory states, separated by the prism according to their differential refrangibility.

Newton's corpuscular theory was not naive. He included a wave-like element to explain periodic phenomena such as the colors of thin films (Newton's rings), introducing what he called "fits of easy reflection and easy transmission" to account for the periodicity. The combined corpuscular-with-fits theory was empirically powerful, accounting for most known optical phenomena of his time, and Newton's authority secured its dominance in English physics for over a century.

The corpuscular theory had genuine virtues. It explained rectilinear propagation directly (corpuscles travel in straight lines unless deflected). It accommodated the discrete arrival of light at detectors (each corpuscle is a discrete unit). It connected naturally to the atomic theory of matter that was beginning to emerge in chemistry. It also had a serious weakness, which Huygens identified and which would prove decisive: it could not account naturally for diffraction, interference, or the failure of two crossed beams of light to scatter each other.

4.2 Huygens and the Wave Theory

Christiaan Huygens, in his *Traité de la Lumière* (1690, written 1678), proposed an alternative. Light is a disturbance propagating through a continuous medium, the luminiferous ether, by means of secondary wavelets. Each point reached by a wavefront becomes the source of new wavelets, and the envelope of these

wavelets constitutes the new wavefront. This *Huygens principle* allowed Huygens to derive rectilinear propagation, reflection, and refraction, including the sine law, from a single mechanism.

Huygens's theory was a return, in refined form, to the Aristotelian picture of light as a state of a continuous medium. The medium (the ether) had to have specific mechanical properties (high stiffness combined with low density, giving high wave speed) to support the observed phenomena. Huygens did not take the ether to be a hypothesis to be added to physics; he took it to be an inference from the wave behavior of light, just as the existence of air is inferred from the propagation of sound.

The wave theory had complementary virtues to the corpuscular. It accounted naturally for diffraction, for the polarization of light (which Huygens identified in calcite), for the failure of crossed beams to scatter, and for the curvature of wavefronts at small obstacles. But it could not, in Huygens's hands, account for the discrete arrival of light at detectors, and it required a medium whose mechanical properties were difficult to reconcile with the apparent emptiness of space.

4.3 The Eighteenth-Century Stalemate

For most of the eighteenth century, the corpuscular theory dominated, supported by Newton's authority and by the genuine difficulty of accounting for rectilinear propagation in a wave theory. The wave theory had its defenders (Euler in particular argued for it), but the empirical case was not yet decisive. The stalemate would last until 1801.

What is striking from the vantage of the present argument is that the corpuscular and wave theories each captured genuine aspects of light's behavior. The corpuscular theory captured the discrete absorption events at detectors. The wave theory captured the continuous propagation through a medium. Each was correct about its own observation and incomplete about the other. The eventual resolution would not be the victory of one over the other but the recognition that both are aspects of a single phenomenon: a continuous-field perturbation registered by quantized matter.

4.4 Young's Double-Slit Experiment

Thomas Young, in a series of papers between 1801 and 1804, presented experiments that revived the wave theory decisively. Young's most famous experiment passed a beam of light through two narrow slits and observed the pattern produced on a screen behind. The pattern was not a uniform brightening, as a corpuscular theory would naturally predict. Instead, it consisted of alternating bright and dark fringes, demonstrating interference.

Interference is the signature behavior of waves. When two waves arrive at the same point in phase, they reinforce; when they arrive out of phase, they cancel. The double-slit pattern is the spatial map of these reinforcements and cancellations. No theory in which light consists of independent particles traveling in straight lines can produce such a pattern, because the particles from one slit cannot cancel the particles from the other.

Young's experiment was initially resisted, partly because of the residual prestige of the Newtonian corpuscular theory and partly because Young's presentation was not yet mathematically rigorous. But the empirical pattern was undeniable, and within a generation it had been confirmed and extended by others, including most decisively Augustin-Jean Fresnel.

4.5 Fresnel's Wave Theory

Fresnel, in a series of papers culminating in his 1818 prize memoir on diffraction, gave the wave theory its mature mathematical form. He showed how Huygens's principle, combined with the principle of interference that Young had identified, could be used to calculate diffraction patterns quantitatively. His theory predicted the famous "Poisson spot" (a bright point at the center of the shadow of a circular obstacle), which Poisson initially derided as absurd and which was then experimentally observed by Arago, confirming Fresnel's theory in a decisive way.

By the 1820s, the wave theory was the consensus position. The corpuscular theory survived only as a historical curiosity. The luminiferous ether was an accepted physical reality (whatever its precise mechanical nature), and light was a transverse wave propagating through it. The continuous medium picture, articulated in pre-formal terms by Aristotle, refined through Alhacen, Descartes, and Huygens, was now the empirical default.

What had been temporarily lost was the corpuscular insight. The discrete arrival of light at detectors had been pushed to the margins, treated as a feature of the detection process rather than as an essential aspect of light. It would return with a vengeance in the early twentieth century.

5. MAXWELL'S SYNTHESIS AND THE ETHER CRISIS

5.1 Electromagnetism Before Maxwell

The first half of the nineteenth century saw the discovery of the principal electromagnetic phenomena: Oersted's discovery (1820) that an electric current produces a magnetic field, Ampère's mathematical theory of the same, Faraday's discovery (1831) of electromagnetic induction, and the slow accumulation of evidence that electric and magnetic phenomena are intertwined. Faraday in particular developed a physical intuition of "lines of force" that propagated through space, an intuition that James Clerk Maxwell would mathematize.

The connection between electromagnetism and optics was suspected before Maxwell. Faraday had shown that strong magnetic fields rotate the polarization of light passing through certain materials (Faraday rotation, 1845). This suggested that light is in some way involved with the same physical entities that produce electric and magnetic phenomena. But the precise relationship awaited Maxwell.

5.2 Maxwell's Equations

James Clerk Maxwell, in "A Dynamical Theory of the Electromagnetic Field" (1865) and in *A Treatise on Electricity and Magnetism* (1873), produced the great synthesis. Maxwell's equations describe the coupled dynamics of electric and magnetic fields, and they have a remarkable consequence: in regions free of charges and currents, the equations admit solutions in which the electric and magnetic fields oscillate transversely, propagating through space at a definite speed. The speed, computed from the electric and magnetic constants of free space, came out to be very nearly the measured speed of light.

Maxwell drew the conclusion. Light is an electromagnetic wave. The luminiferous ether of the wave theorists is the same as the medium that supports electric and magnetic phenomena. Optics is a chapter of electromagnetism.

The unification was extraordinary. It explained why light has the specific speed it does (the speed is determined by the electromagnetic properties of the medium). It explained why light is transverse (the electric and magnetic fields oscillate perpendicular to the propagation direction). It explained Faraday rotation, polarization, and the broad outlines of optical phenomena, all from a single set of equations. Heinrich Hertz's experimental production and detection of radio waves in 1887-1888 confirmed Maxwell's prediction directly: electromagnetic radiation exists at frequencies below the visible, behaves as Maxwell's equations predict, and propagates at the speed of light.

5.3 The Spectrum Extends

The Maxwellian framework extended the spectrum of light far beyond the visible. Infrared (Herschel, 1800) and ultraviolet (Ritter, 1801) had already been recognized as invisible extensions of the visible spectrum, sharing its essential properties. Maxwell's framework explained what those extensions were: electromagnetic waves at frequencies above and below the visible band. Hertz's radio waves were further extensions at much lower frequencies. X-rays (discovered by Röntgen in 1895) and gamma rays (identified in radioactive decays around the turn of the twentieth century) extended the spectrum upward at frequencies vastly higher than the visible.

The spectrum, on the Maxwellian view, is a single continuous range of electromagnetic radiation, parameterized by frequency. All of it propagates at the same speed in vacuum, the speed c . The differences across the spectrum are differences of frequency and wavelength, not differences of underlying nature. A radio wave and a gamma ray differ in oscillation rate, but they are perturbations of the same field, governed by the same equations, traveling at the same speed.

This is a structural point that will matter in Section 8, where the relationship between propagation speed and other physical quantities is examined more carefully. For the moment, the key observation is that the Maxwellian framework gives the unification of the spectrum its proper grounding: the substrate (the electromagnetic field) is one, and its perturbations differ in oscillation rate without differing in their fundamental kinematic properties.

5.4 The Ether Crisis

Maxwell's equations described the propagation of electromagnetic waves through a medium, the ether. The ether had to have peculiar mechanical properties. It had to be stiff enough to support transverse waves at the speed of light, but it had to be so rarefied that the planets and other massive bodies could move through it without measurable resistance. Various models were proposed (Maxwell himself entertained mechanical analogies involving cells and rotating fluids), but none was entirely satisfactory.

Worse, the ether, on the Maxwellian view, defined a preferred frame of reference. Light should propagate at speed c relative to the ether, and the motion of the Earth through the ether should produce measurable effects on the speed of light as observed in different directions. Albert Michelson and Edward Morley, in their famous 1887 experiment, set out to detect this motion. Using an interferometer of unprecedented sensitivity, they searched for the predicted directional dependence of light's speed.

They found nothing. Within the precision of the experiment, the speed of light was the same in all directions, regardless of the Earth's motion. The result was

repeated and refined over the following decades, always with the same outcome. The Earth, it appeared, was not moving through the ether.

The Michelson-Morley result created a crisis. Either the ether was somehow being dragged along by the Earth (a position defended by Stokes and others, but increasingly difficult to reconcile with observations of stellar aberration), or the laws of electromagnetism took the same form in every reference frame (which seemed to require abandoning the ether as a meaningful physical entity), or some more radical reconception was needed.

The resolution would come from Albert Einstein in 1905, but it would be a resolution that went deeper than anyone in the nineteenth century had anticipated. The ether would not be patched. It would be reconceptualized.

5.5 The Lesson of the Ether Crisis

The lesson of the ether crisis, often misstated, is not that the ether was abandoned. The lesson is that the ether, as the nineteenth century conceived it, was a hybrid of two ideas that needed to be separated: the idea of a medium for electromagnetic propagation, and the idea of a fixed mechanical substance defining a kinematically detectable preferred frame. The first idea is defensible and survives in modern physics. The second is empirically refuted by Michelson-Morley and its successors.

What modern physics offers in place of the mechanical ether are two distinct interpretive options that make identical empirical predictions but differ in ontology. The first is the standard reading of special relativity: Lorentz-invariance is primitive, no preferred frame exists, and the apparent absence of a preferred frame is fundamental rather than emergent. The second is the Lorentz-Poincaré ether tradition, refined by modern thinkers including Bell, Maudlin, and others: the substrate is physically real, kinematic experiments cannot distinguish its rest frame from any other inertial frame because Lorentz contractions and time dilations conspire to produce frame-symmetric measurements, but the substrate is ontologically prior to the kinematic appearances.

These two positions are empirically equivalent in all currently tested regimes. They differ on whether the absence of a preferred frame is a metaphysical claim about reality (standard SR) or an emergent feature of substrate dynamics (neo-Lorentzian). The choice between them is a philosophical question rather than an experimental one in the present epoch.

The present paper takes the neo-Lorentzian option openly. The substrate is treated as physically real and continuous. Lorentz-invariance of the field equations holds as an emergent kinematic feature ensuring no preferred frame is detectable by any local experiment, while leaving open the ontological status of the substrate's rest frame as a real but kinematically inaccessible structure.

This commitment is consistent with Bell's late writings on the question, with Maudlin's defense of Lorentzian relativity, and with a substantial minority tradition in the philosophy of physics. It is empirically equivalent to standard SR in every tested regime; it differs only in interpretive flavor, and the differences become salient when the framework is asked questions that involve the substrate's substantive ontology (entanglement, cosmological expansion, the nature of time dilation).

The medium is real but is not a body. The medium is what we call space when we ask about its electromagnetic properties. The propagation of perturbations occurs through this medium at a definite speed c . The substrate's own dynamics are constrained such that no kinematic measurement reveals its rest frame, but the substrate is no less real for that. The continuity of the substrate, the requirement of a medium for propagation, the impossibility of the void as the locus of optical phenomena: these survive into modern physics under this interpretive option, properly articulated.

6. THE TWENTIETH-CENTURY REVOLUTION

6.1 Einstein's Photoelectric Paper

In 1905, Einstein published four papers of extraordinary importance, three of which bear directly on the nature of light. The first, "On a Heuristic Viewpoint Concerning the Production and Transformation of Light" (March 1905), reintroduced the corpuscular aspect of light that the wave theory had displaced.

The photoelectric effect had been studied since the 1880s. When light strikes a metal surface, electrons are sometimes ejected. The behavior is peculiar: the ejection of electrons depends on the frequency of the light, not on its intensity. Below a threshold frequency, no electrons are ejected, no matter how intense the light. Above the threshold, electrons are ejected immediately, and their kinetic energy increases linearly with the frequency of the light, again independent of intensity.

The wave theory of light could not account for this. On a wave theory, the energy of a light wave should depend on its intensity, and electrons should be ejected when sufficient energy has accumulated, which would mean a delay at low intensities. The observed instantaneous ejection above threshold, and the dependence on frequency rather than intensity, suggested that light delivers its energy in discrete packets whose energy is proportional to frequency.

Einstein's paper proposed exactly that. Light, he argued, behaves as if it consisted of discrete quanta of energy, each carrying energy E equal to h times the frequency, where h is Planck's constant (introduced by Planck in 1900 to fit the black-body spectrum). The photoelectric effect is then explained: each quantum delivers its energy to a single electron; if the quantum's energy is below the threshold (the work function of the metal), no electron is ejected; if above, the excess energy becomes the electron's kinetic energy.

Einstein was awarded the Nobel Prize in 1921 for this work. The corpuscular aspect of light, which had seemed dead, was alive again. But it was alive in a peculiar form. Light still showed wave behavior in interference experiments. It also showed particle behavior in the photoelectric effect. The two aspects appeared incompatible at first sight, and the apparent incompatibility became the central interpretive problem of twentieth-century physics.

6.2 Special Relativity

Einstein's third 1905 paper, "On the Electrodynamics of Moving Bodies" (June 1905), founded special relativity. The theory begins from two postulates: the laws of physics take the same form in every inertial reference frame, and the speed of

light in vacuum is the same constant c in every inertial reference frame, regardless of the motion of the source.

The constancy of c follows directly from Maxwell's equations being the same in every frame. The two postulates together force a reconception of space and time. The Galilean assumptions about how space and time relate across frames must be abandoned, and the Lorentz transformations adopted in their place.

Special relativity makes a number of empirical predictions that have all been confirmed: time dilation (clocks moving relative to an observer run slow), length contraction (objects moving relative to an observer are shortened in the direction of motion), the relativity of simultaneity (events simultaneous in one frame are not simultaneous in another), and the equivalence of mass and energy ($E = mc^2$). The theory has been verified to extraordinary precision in particle accelerators, in the navigation of spacecraft, in the synchronization of GPS satellites, and in the lifetime measurements of unstable particles.

Special relativity does not, in itself, tell us what light is. It tells us how light's propagation speed relates to the kinematics of observers and how electromagnetic phenomena transform between frames. The interpretive question (what is light, ontologically?) is left open. Einstein's photoelectric paper had reintroduced the corpuscular aspect; his relativity paper had given the kinematics of electromagnetic propagation an unexpectedly elegant form. The two papers together set the agenda for the next half-century of physics.

6.3 Wave-Particle Duality

The decade after Einstein's photoelectric paper saw further confirmation of light's particle aspect. The Compton effect (1923), in which X-rays scatter from electrons with a wavelength shift proportional to the scattering angle, was decisive: light carries momentum as well as energy in discrete packets, and the scattering can be analyzed as a relativistic collision between a photon and an electron. The photon was now, in effect, a particle.

But the wave behavior persisted. Diffraction and interference experiments continued to show the patterns predicted by Maxwell's wave theory. The double-slit experiment, performed with single photons one at a time, showed that the interference pattern builds up gradually as photons accumulate on the detector. Each photon arrives at a definite point (particle behavior), but the distribution of arrival points follows the wave-theoretic interference pattern (wave behavior). Each individual photon, in some sense, "passes through both slits" and interferes with itself.

This is the famous wave-particle duality. By the late 1920s, with the establishment of quantum mechanics by Heisenberg, Schrödinger, Born, Dirac, and others, duality was generalized: every physical entity, not only light, exhibits

both wave and particle aspects. The aspects are not contradictory but complementary, in Niels Bohr's formulation. The aspect that appears depends on the experimental arrangement.

The Copenhagen interpretation, dominant in the mid-twentieth century, treated this duality as a fundamental feature of nature, beyond which physical theory could not go. The wave function of an entity describes its propagation; measurement collapses the wave function and yields a definite outcome. Why measurement does this, and what the wave function "really is," were questions the Copenhagen interpretation declined to answer.

6.4 Quantum Field Theory

The deeper resolution of wave-particle duality came with quantum field theory. Beginning with Dirac's 1927 quantization of the electromagnetic field and developing through the work of Heisenberg, Pauli, Fermi, Tomonaga, Schwinger, Feynman, Dyson, and many others, quantum field theory recasts the basic ontology of physics.

In quantum field theory, the fundamental entities are fields, not particles. The electromagnetic field is one such field. The field exists everywhere in space; it has a quantum state at every point; and its excitations are what we call photons. A photon is not a tiny particle. A photon is a localized excitation of the electromagnetic field, characterized by an energy, a momentum, and a polarization. The same field can have many photons, or none, or be in superpositions of states with different photon numbers.

Wave-particle duality, on this reading, is not a paradox. It is a consequence of the field-excitation structure. The "wave" aspect is the propagation of the field excitation according to the field equations. The "particle" aspect is the discrete way in which the field interacts with matter: emission and absorption events involve definite quanta of energy and momentum because the field's interactions with matter are quantized at the level of the matter (atomic transitions are discrete) as well as the field (each quantum carries a definite amount).

This is essentially the substrate-perturbation view that the present paper articulates. The field is the substrate. Its excitations are the perturbations. The substrate is continuous; the perturbations propagate as waves; the interactions with matter are quantized because matter has discrete states. Aristotle's actualization of the transparent, Huygens's wave in the ether, Maxwell's electromagnetic wave, and the modern photon are not four different theories; they are four formulations of the same geometric structure, with increasing mathematical precision and empirical scope.

7. THE SUBSTRATE-PERTURBATION READING STATED

The view that emerges from the historical sweep, and that quantum field theory has effectively confirmed, can be stated directly.

Light is a perturbation propagating through a continuous substrate (in the framework's vocabulary, the Impressed Plenum, identified with the electromagnetic field considered as a physically real entity rather than merely a mathematical convenience). The photon, on this reading, is a quantum excitation of the substrate. The excitation is not a tiny particle traveling along a definite path; it is a localizable disturbance of the field whose propagation follows the field equations and whose interactions with matter are quantized at the level of energy and momentum exchange. This is the standard quantum-field-theoretic ontology, articulated with emphasis on the substrate-as-real component.

The framework's "knot" or "topological perturbation" terminology applies most naturally to massive solitonic excitations (where appropriate topological structure can be defined and where rest mass arises from the topological charge of the configuration), and is used cautiously and in specific contexts. Photons are not topological solitons in standard quantum electrodynamics; they are perturbative excitations of a free gauge field. The paper retains "perturbation" as the general descriptor and reserves "knot" for cases where the topological structure does the explanatory work, such as in the Faddeev-Niemi-style Hopf-soliton models for fermion mass hierarchies that the framework explores elsewhere. For the photon specifically, the descriptor is "field excitation" or "perturbation of the substrate."

The substrate is not a thing within space. The substrate is space, considered with respect to its electromagnetic properties.

The substrate has its own structure, including a definite intrinsic relaxation rate, which determines the propagation speed of perturbations within it. That speed is c , the speed of light in vacuum. It is not the speed of any particular object; it is the speed at which a perturbation of the substrate propagates from one location to another. It is, equivalently, the speed at which causal influence can propagate through the substrate.

The substrate also has a bodily extension. The substrate can grow, stretch, or contract as a whole, independently of any propagation occurring within it. This is not an additional motion of objects in space; it is a change in the substrate itself. The cosmological metric expansion, established empirically since the 1920s and given precise quantitative form in modern cosmology, is exactly this kind of bodily extension. The substrate's expansion rate is not constrained by c , because c is the rate of perturbations within the substrate, not the rate of the substrate itself. A wave on a string cannot exceed the speed of waves on the string, but the

string itself can be stretched at any rate, including rates that exceed the wave speed.

Perturbations of the substrate come in many forms. Massless excitations (the photons of the electromagnetic field, the gravitons of the gravitational field if they exist) propagate at c . Massive excitations (the quanta of fields with non-zero rest mass: the W and Z bosons of the weak interaction, the Higgs boson, and via field excitations the leptons and quarks) propagate at speeds less than c , governed by their rest mass. The hierarchy of speeds, properly stated, is determined by rest mass: zero rest mass corresponds to propagation at c ; non-zero rest mass corresponds to slower propagation, with heavier excitations slower than lighter ones.

Frequency, contrary to a common misconception, is not on the same axis as rest mass. Two photons of different frequencies (a radio photon and a gamma photon, say) have the same rest mass (zero) and propagate at the same speed (c). They differ in oscillation rate, energy, and momentum, but not in speed. This is the essential correction that Section 8 will develop: frequency is orthogonal to the rest-mass-dependent propagation hierarchy.

Discrete absorption events at detectors are a property of the detector, not of the field. The substrate's perturbations are continuous; the matter that detects them has discrete states; the interaction between continuous field and discrete matter produces discrete events. A photoelectron is ejected because the absorbing atom can only accept a quantum of energy that matches a transition in its electronic structure. The light didn't arrive in a packet because the light is a packet; the light arrived in a packet because the atom can only swallow packets.

This reading does not contradict quantum field theory. It is the geometric content of quantum field theory, articulated in language that emphasizes the substrate-perturbation structure rather than the operator-algebraic formalism. Quantum field theorists who think carefully about what their equations say are saying essentially the same thing, sometimes in different words. The framework's contribution is to insist on this geometric reading explicitly and to show that it dissolves several interpretive paradoxes that the operator-algebraic formalism leaves open.

8. THE HIERARCHY OF SPEEDS PROPERLY STATED

A frequently encountered error, including in some preparatory drafts of the present paper, holds that high-frequency electromagnetic radiation (gamma rays) propagates more slowly than low-frequency radiation (radio waves), or that the propagation speed of light depends in some way on its frequency. This view is empirically false and structurally confused. It must be corrected explicitly, because the substrate-perturbation reading is otherwise vulnerable to dismissal on this point alone.

8.1 The Empirical Constraint

Multi-messenger astronomy provides the decisive empirical evidence. On August 17, 2017, the LIGO and Virgo detectors recorded gravitational waves from the merger of two neutron stars, an event designated GW170817. Within 1.7 seconds of the gravitational-wave signal's end, the Fermi Gamma-ray Burst Monitor and INTEGRAL detected a short gamma-ray burst from the same direction. Subsequent observations across the full electromagnetic spectrum (radio, optical, infrared, ultraviolet, X-ray) detected the kilonova counterpart over the following weeks.

The neutron-star merger occurred approximately 130 million light-years from Earth. Gravitational waves and gamma rays from the event arrived within about 1.7 seconds of each other, after traveling for 130 million years. This constrains the speed difference between gravitational waves and gamma rays to roughly one part in 10^{15} . By the same logic, comparing gamma rays with the lower-energy electromagnetic emission detected over the days following the event, the speed of electromagnetic radiation is the same across the spectrum to extraordinary precision.

This is not a theoretical constraint. It is a measured fact. Whatever the framework says about the structure of light, it cannot say that gamma rays travel at a different speed in vacuum from radio waves. They travel at the same speed, c , to a precision that has been empirically verified across the entire observable spectrum.

8.2 The Structural Reason

The structural reason is straightforward, once stated. Within quantum field theory, the photon is the massless excitation of the electromagnetic field. All photons, regardless of frequency, are excitations of the same field with the same rest mass (zero). The propagation speed of a massless excitation in a Lorentz-

invariant theory is c , the universal speed of causality, which is set by the geometry of spacetime itself.

Different frequencies correspond to different oscillation rates of the excitation, not to different kinds of excitations. The energy and momentum of a photon scale with frequency ($E = hf$, $p = hf/c$), but the propagation speed does not. The frequency parameterizes the internal oscillation, not the external translation.

In the framework's substrate-perturbation language: knot-density (the topological complexity that determines rest mass) is one quantity; oscillation rate (the frequency of the excitation) is another. A radio photon and a gamma photon have the same knot-density (zero, since both are massless); they differ only in oscillation rate. They are siblings, not parent-and-child in the propagation hierarchy.

The hierarchy of propagation speeds, properly stated, is:

1. The substrate's bodily extension (metric expansion of space). Not constrained by c . Can occur at any rate, including rates much greater than c , as during cosmological inflation or in the present-day expansion of the universe at scales beyond the Hubble horizon.
2. Massless field excitations. All propagate at c . This includes all electromagnetic radiation regardless of frequency (radio, microwave, infrared, visible, ultraviolet, X-ray, gamma). It also includes gravitational waves and any other massless field excitations.
3. Massive field excitations. Propagate at speeds less than c , with the maximum speed approached as the kinetic energy becomes large compared to the rest energy. The hierarchy among massive excitations is determined by rest mass: lighter excitations (electrons, light leptons) reach higher speeds at given energies than heavier excitations (protons, alpha particles, heavy nuclei) at the same kinetic energy.
4. Macroscopic matter. Aggregate motion of bound systems. Always less than c ; in practice, much less, because the rest energy of macroscopic matter is enormous and accelerating it to relativistic speeds is energetically prohibitive.
5. Rest, in some local frame of reference.

Frequency does not enter this hierarchy. A radio photon and a gamma photon both occupy slot 2. An electron and an alpha particle both occupy slot 3, with the electron above the alpha particle in the within-slot ordering by virtue of its lower rest mass.

8.3 Why the Misconception Recurs

The misconception that high-frequency electromagnetic radiation might be slower than low-frequency arises from a natural but incorrect intuition. High-frequency radiation carries more energy per quantum and produces more dramatic interactions with matter (gamma rays ionize; radio waves do not). It is tempting to associate this energetic intensity with greater "thermodynamic mass" and to suppose that greater thermodynamic mass implies slower propagation.

The temptation must be resisted. Energy is not mass, in the relevant sense. The rest mass of a photon is zero, regardless of the photon's energy. The energy of a photon is its kinetic energy, in effect, which determines what interactions it can produce but does not affect its kinematic propagation. The substrate couples to all massless excitations identically; what varies is the oscillation rate of the excitation, not its propagation through the substrate.

A useful intuition: imagine a long elastic string under uniform tension. Waves of different wavelengths can travel along the string. Short wavelengths (high frequencies) and long wavelengths (low frequencies) all travel at the same wave speed, which is determined by the tension and density of the string, not by the wavelength of the particular wave. A high-frequency wave wiggles more rapidly within itself, but it does not travel along the string any faster or slower than a low-frequency wave. The string's properties determine the propagation speed; the wave's frequency determines its internal oscillation rate.

This is exactly the situation with electromagnetic radiation. The substrate's properties determine the propagation speed (c). The radiation's frequency determines its oscillation rate, which is orthogonal to its propagation through the substrate.

9. MODERN PHENOMENA REVISITED

The substrate-perturbation reading must accommodate, and indeed illuminate, the full range of modern optical and quantum phenomena. This section treats the most important.

9.1 The Photoelectric Effect

A continuous electromagnetic perturbation strikes a metal surface. The atoms of the metal have discrete electronic states. An electron can absorb energy from the perturbation only in amounts matching its available transitions, and only enough at once to overcome the work function. Below threshold frequency, the energy per quantum is too small to liberate any electron, no matter how intense the radiation. Above threshold, each absorbed quantum delivers enough energy to liberate one electron, with the excess becoming kinetic energy. The kinetic energy depends on frequency (since each quantum's energy is hf) and not on intensity (since intensity affects the number of quanta arriving per second, not the energy each one delivers).

The photoelectric effect, on the substrate reading, is a consequence of the discrete atomic structure of matter, not of any discrete structure of the field. The field is continuous; matter is quantized; the interaction therefore proceeds in discrete events whose energy is set by the matter's transition energies.

The point is critical. Einstein's 1905 paper has often been read as establishing that light "is" particles, that the field is fundamentally discrete. The substrate reading reverses the inference. Light's interactions with matter are discrete because matter is discrete. The light itself, as a perturbation of the field, remains continuous in its propagation. The discrete events at detectors are events of the detector, not events of the light.

9.2 The Double-Slit Experiment

A continuous wave of electromagnetic perturbation passes through two parallel slits. Each slit becomes a secondary source (Huygens's principle). The wavefronts from the two secondary sources overlap and interfere. Where the path lengths from the two slits differ by an integer number of wavelengths, the wavefronts reinforce; where they differ by half-integer wavelengths, the wavefronts cancel. The result is a pattern of bright and dark fringes on the screen behind the slits.

This is the wave-theoretic account, and it is correct. What was puzzling, on the corpuscular reading of the photon, is that the same pattern emerges when photons are sent through one at a time. Each individual photon arrives at a definite point on the screen (a discrete event, governed by the absorption

properties of the screen's atoms). But the cumulative distribution of arrival points, as more and more photons accumulate, is exactly the wave-theoretic interference pattern. This is the central interpretive puzzle of quantum mechanics, often dramatized as "each photon interferes with itself" or "the photon goes through both slits."

The substrate reading dissolves the puzzle. The photon is a localized excitation of the electromagnetic field. The excitation is not a tiny particle that travels along a definite path. The excitation is a distributed perturbation of the field, which propagates as the field equations specify (passing through both slits, interfering with itself). The discrete arrival event is the absorption of the excitation by a single atom in the screen; the location of that absorption is determined probabilistically by the field's amplitude at each point (since the field's amplitude squared is the probability density for absorption). Many photons accumulate the interference pattern because each individual absorption is a probabilistic sampling of the same underlying field distribution.

The wave goes through both slits because the wave is the field, and the field is everywhere. The particle arrives at one point because the absorption is a single discrete event. There is no contradiction. The wave aspect and the particle aspect are aspects of the field-matter interaction, not aspects of the field alone or the matter alone. The continuous field passes through both slits and interferes; the discrete matter at the detector absorbs at one location.

This is the same point as Section 9.1. Discrete measurement, continuous field; the discreteness is at the matter, the continuity is at the field; the interaction is the place where they meet.

9.3 Quantum Entanglement and Bell-Test Correlations

Two photons can be prepared in a quantum state such that their individual properties (polarizations, say) are correlated even when they are spatially separated. Measurements of one photon's polarization yield results that are statistically correlated with measurements of the other photon's polarization, even when the measurement events are space-like separated (so that no light-speed signal could connect them).

Bell's theorem (1964) shows that the observed correlations cannot be explained by any theory in which each photon has definite, locally-determined properties before measurement. Aspect's experiments (1982) and many subsequent refinements have confirmed the violation of Bell inequalities to overwhelming statistical significance. Entanglement is real, and it is genuinely non-classical.

The orthodox interpretation treats entanglement as a fundamental feature of quantum mechanics that resists straightforward classical or local-realistic description. Various interpretive moves have been proposed (many-worlds,

hidden variables of the non-local kind, retrocausal accounts, QBism) but no consensus has emerged.

The substrate reading offers an interpretive frame consistent with the empirical findings, though it must be honest about its limits. Two entangled photons share a single non-separable field-state. The substrate is one continuous medium; the field-state is one quantum state of that medium; the spatial separation between the two registration events is a feature of the registration, not of the field's quantum state. The framework's continuous-field ontology is consistent with this picture. It is not, however, a derivation of the specific Bell correlations from substrate first principles in any way that goes beyond what standard quantum field theory already says. The framework restates the entanglement structure in geometric language; it does not derive the cosine-squared form of the correlations or predict them from substrate dynamics in a manner standard QFT cannot.

What the framework adds is interpretive clarity. The substrate is real and continuous; the field-state on it is non-separable; the registration events are local interactions between the field and discrete matter. There is no separate question of "how the photons communicate" because there are no separate photons in the substrate-mode of description; there is one field-state being sampled at two locations. This dissolves the appearance of action-at-a-distance while leaving the empirical content of Bell's theorem intact. It does not, and does not claim to, derive Bell correlations from substrate ontology in a way that constitutes new physics.

For readers seeking a fully derivational account of why specific Bell correlations take the values they do, standard quantum field theory provides the calculations. The substrate reading is interpretive scaffolding around those calculations, not a replacement for them. Honest acknowledgment of this limit strengthens rather than weakens the framework's interpretive contribution.

9.4 Redshift in Three Flavors

When light from a distant source arrives with wavelengths longer than the wavelengths it was emitted with, the phenomenon is called redshift. Three distinct mechanisms produce redshift, and they must be distinguished.

Doppler redshift occurs when the source is moving away from the observer through the substrate. The wavefronts emitted later originate from positions farther from the observer than the wavefronts emitted earlier, so the time between successive wavefronts at the observer is longer than the time between their emissions, and the wavelength as observed is correspondingly longer. Doppler redshift is a kinematic effect dependent on the relative motion of source and observer.

Gravitational redshift occurs when light propagates from a region of stronger gravitational potential to a region of weaker gravitational potential (climbing out of a gravity well). The photon's energy decreases as it does work against the gravitational gradient, and since photon energy is proportional to frequency ($E = hf$), the frequency decreases and the wavelength lengthens. Gravitational redshift is an effect of substrate tension (in the framework's reading) or of metric structure (in the standard relativistic reading).

Cosmological redshift is the redshift of light from distant galaxies, increasing systematically with distance. It is not the Doppler effect (the receding galaxies are not, in general, moving through the substrate; the substrate itself is expanding between us and them). It is the stretching of the photon's wavelength by the bodily expansion of the substrate during the photon's travel. As the substrate expands, the wavelength of any propagating perturbation expands with it. The redshift, in this case, is a direct measure of the substrate's expansion since the photon was emitted.

Cosmological redshift is critical for the substrate reading. It demonstrates empirically that the substrate has an independent bodily extension (metric expansion) that is distinct from any motion within the substrate. The substrate is not a passive frame; it is a dynamic medium whose own dimensions change over time. The redshift of distant galaxies is the photographic evidence of that change.

9.5 Time Dilation and the Twin Paradox

A clock moving at high speed relative to an observer ticks slow as seen by the observer. This is time dilation, predicted by special relativity and confirmed empirically (in the lifetime of fast-moving muons, in the behavior of atomic clocks on aircraft and satellites, in the navigation corrections required for GPS, and in countless particle-physics experiments).

The famous twin paradox concerns two twins, one of whom remains at home while the other travels at high speed and returns. The traveling twin returns younger than the stationary twin. The asymmetry is real and has been verified, in idealized form, in clock-based experiments.

Two interpretive options handle this empirical fact equivalently. The standard reading of special relativity treats time dilation as an effect of spacetime geometry: time is the fourth dimension, motion through space affects the passage of time as a geometric consequence of the Lorentzian metric, and the asymmetry between twins is explained by the traveling twin traversing a different path through spacetime, with shorter proper time. This is mathematically clean and empirically correct.

The neo-Lorentzian reading the present paper develops treats time dilation as a feature of substrate-engagement. A clock moving with respect to the substrate's rest frame undergoes physical effects on its internal processes that produce the empirical signature of time dilation. The asymmetry between twins, on this reading, requires specifying which twin moves at higher substrate-relative velocity during the inertial segments. This specification is not arbitrary, and the framework owes a clear answer.

The natural identification, in cosmological contexts, is that the substrate's rest frame coincides with the cosmic microwave background rest frame (the frame in which the CMB dipole vanishes). This is a frame globally privileged by the universe's matter-energy distribution; Earth has a known peculiar velocity of approximately 370 km/s relative to it. For purely local analysis of a twin-paradox configuration, however, the framework can also adopt the original parting frame as the local reference, with the understanding that this is a local convenience consistent with the global CMB-frame identification at cosmological scales. Both choices give identical empirical predictions because Lorentz invariance ensures local kinematic experiments cannot distinguish among inertial frames.

With the substrate rest frame thus specified (CMB-frame globally, parting-frame locally), the substrate-relative velocity of each twin is well-defined throughout the experiment. The cycle-count accumulated by each twin between parting and reunion is the integral of his substrate-relative time element along his worldline, and this integral yields a smaller value for the worldline that includes a high-substrate-velocity traveling segment. The asymmetry between twins is settled at the level of who underwent the acceleration that took him to high substrate-relative velocity. Acceleration breaks the twin symmetry in any account, including standard SR; what the neo-Lorentzian reading adds is the physical mechanism (substrate-engagement during the high-velocity segment) by which the cycle-count differential arises.

The local kinematic invariance of physics ensures that no local experiment can detect the substrate's rest frame from a single inertial reference. This is consistent with the empirical content of special relativity; the framework's commitment is that this kinematic invariance is an emergent consequence of the substrate's dynamics rather than a metaphysical primitive about the absence of any rest frame whatsoever.

Both interpretations make identical empirical predictions. The choice between them is philosophical. The neo-Lorentzian reading takes on a specific interpretive cost: it commits to a real (though kinematically undetectable in any local inertial experiment) substrate rest frame. It earns a corresponding interpretive benefit: it provides a concrete physical mechanism (substrate-engagement of internal processes) for what the standard reading treats as a pure geometric fact (different proper times along different worldlines).

A clarification is required to address an apparent circularity in the substrate-engagement language. To say one clock "moves through the substrate" and another does not might seem to require a preferred frame from which the comparison is made, and to thereby contradict the empirical absence of any frame-dependent effect detectable in local experiments. The resolution is that the comparison between the twins is not made in any third-party frame; it is made by the twins themselves at their reunion, by counting internal cycles accumulated between two events they both share (parting and reunion). Each twin's count is a fact about his own worldline, not about his relation to any external time. The traveling twin counts fewer cycles between the shared events because his worldline through spacetime (or equivalently, his pattern of substrate-engagement during travel) accumulates fewer of his own internal changes per unit of shared interval. No external time is required for this comparison; only the shared boundary events and the internal cycle-counts.

What the neo-Lorentzian reading adds, beyond what standard SR provides, is an account of why the cycle-counts differ. The standard reading says they differ because the worldlines differ in proper time, which is a primitive geometric quantity. The neo-Lorentzian reading says they differ because the traveling twin's substrate-engagement during the traveling phase produces physical slowing of his internal processes. Both accounts arrive at the same observable difference. The neo-Lorentzian account is more committed ontologically; the standard account is more parsimonious. Neither is empirically distinguished from the other in tested regimes, but each carries different implications for how related questions (cosmological metric expansion, the substrate's role in gravity, the interpretation of entanglement) are handled.

The traveling twin returns younger because his physical processes accumulated fewer cycles between parting and reunion than his brother's did. He aged less because he did less, in the sense of internal change. Time as such did not warp for him; what changed was the rate of process accumulation in his localized substrate region during the traveling phase. The dilation is real and observable; it is a dilation of process rates, not of an external time.

9.6 Cosmological Inflation and Metric Expansion

In modern cosmology, the universe underwent a period of extremely rapid expansion in its earliest moments (cosmological inflation, lasting roughly 10^{-35} to 10^{-32} seconds after the initial singularity), during which spatial dimensions expanded by a factor of approximately 10^{26} or more. After this inflationary epoch, the expansion continued at a slower but still significant rate, governed in the present epoch by dark energy and matter densities.

At sufficient distances, the recession velocity of distant galaxies, computed from cosmological redshift, exceeds c . This is not a violation of relativity. The galaxies are not moving through the substrate at superluminal speeds; the substrate itself

is expanding between us and them, and the recession we observe is the cumulative effect of metric expansion over the photon's travel time.

This is the cleanest empirical demonstration of the substrate-perturbation distinction that the present paper urges. The substrate has its own bodily extension. That extension is not bounded by c , because c is the rate of perturbations within the substrate, not the rate of the substrate itself. The substrate can expand at any rate; what it cannot do is propagate causal influence faster than c , because c is precisely the rate at which causal influence propagates through the substrate.

This is the distinction captured by the elastic-string analogy: the guitar string under tension can be stretched at any rate, but waves on the string can only travel at the wave speed determined by the string's tension and density. The wave is bounded by the string's properties; the string itself is bounded by whatever determines its overall extension, which is a separate question.

Standard cosmology accepts this distinction without difficulty. The substrate-perturbation reading grounds the distinction ontologically: the substrate is a physical entity with its own dynamics, distinct from the dynamics of perturbations within it. What the substrate is, on this reading, is what space is; what the substrate's extension is, is what cosmological expansion is; the two are not separate questions.

9.7 Gravitational Lensing

When light passes near a massive object, its path is deflected. The deflection was predicted by Einstein's general relativity and famously confirmed by Eddington's observations of stars near the Sun during the 1919 eclipse. Modern observations of gravitational lensing (the bending of light by galaxies and galaxy clusters, producing multiple images, arcs, and rings of distant background sources) are routine in astronomy.

The orthodox general-relativistic interpretation treats lensing as a consequence of spacetime curvature. Mass curves the spacetime around it; light follows null geodesics through the curved spacetime; the apparent bending is the geodesic structure. This is mathematically clean and empirically accurate.

The substrate reading reframes the same phenomenon in different ontological terms. Mass produces a tensional gradient in the substrate; light, as a perturbation propagating through the substrate, follows the path of least resistance through the gradient; the path of least resistance is bent toward the mass. The empirical predictions are the same; the underlying picture differs.

Two observations support the substrate reading as a useful complement to the geometric reading. First, the substrate reading makes the role of the substrate explicit: light bends because the medium through which it propagates has

variable properties induced by mass. Second, the substrate reading connects gravitational lensing naturally to the other phenomena discussed here (gravitational redshift, time dilation, cosmological expansion), all of which are then aspects of the substrate's structure and dynamics. The geometric reading also unifies these phenomena, but only through the abstract metric tensor; the substrate reading gives the unification a physical referent.

9.8 Black-Hole Event Horizons

A black hole is a region of spacetime from which no signal can reach an external observer. The boundary of this region is the event horizon. In standard general relativity, the event horizon is a global feature of the spacetime geometry rather than a local one. There is no curvature singularity at the horizon, no local tension extremum, no physically distinguished property of the substrate at the horizon's location. An observer falling through the horizon experiences nothing locally remarkable at the moment of crossing. The horizon is identifiable only globally: it is the boundary beyond which all future-directed timelike paths lead inward to the central singularity, with no return path available to the exterior.

This non-local character of the horizon must be respected by any interpretive reading. An earlier draft of this paper described the horizon as a coordinate where "substrate tension" prevents outward propagation. That formulation is misleading because it suggests something locally extreme happens at the horizon. The corrected substrate reading treats the horizon as a global feature of the substrate's causal structure rather than a local field-strength extremum. The substrate's geometric configuration in the region surrounding the black hole is such that the future light cones tip progressively toward the central mass, until at the horizon's location the future light cones tip entirely inward and outward propagation is no longer geometrically available. This tipping is a global property of the geometry; it is not localized at the horizon's coordinate.

The substrate reading and the standard general-relativistic reading agree on this picture. Both describe the horizon as a global causal-structure feature. Both predict the same observational signatures (gravitational lensing around the black hole, frequency shift of light escaping from near the horizon, eventual Hawking radiation, the impossibility of receiving any signal from inside). The interpretive difference is whether the global causal structure is a feature of the metric tensor (standard reading) or of the substrate's geometric configuration (substrate reading). The two descriptions are mathematically equivalent in the regimes where general relativity has been tested.

What the substrate reading does not do, and should not pretend to do, is provide local-mechanical pictures (such as "tension extremum") that fail to correspond to the actual geometry. The horizon is a global feature; any reading that localizes it to a local field-strength is mistaken. The substrate reading is not committed to

any such localization, and the present paper now states the global character explicitly.

For the framework's broader claim of unification across optical phenomena, the horizon is one more aspect of the substrate's geometric structure rather than a local property. Light bends in the substrate's gradient (lensing); photons climb out of the gradient with shifted frequency (gravitational redshift); the global geometry traps light beyond a certain coordinate (event horizon). These are aspects of one substrate-geometric dynamic, treated here in the same vocabulary as in standard general relativity but with the substrate ontology made explicit. The unification is real; the interpretive cost is the substrate-realism commitment that the standard reading does not require.

9.9 The Spectrum from Radio to Gamma

The full electromagnetic spectrum, from radio through microwave, infrared, visible, ultraviolet, X-ray, and gamma, consists of perturbations of a single field at different frequencies. All of these perturbations are massless excitations of the electromagnetic field (photons of zero rest mass). All propagate at c in vacuum, regardless of frequency, as confirmed by multi-messenger astronomy.

The differences across the spectrum are differences of frequency and consequent differences of energy per quantum, with corresponding differences in how the radiation interacts with matter. Radio waves are absorbed and emitted by free or weakly-bound electrons in conductors; their photons have energies far below typical atomic transitions, so they do not produce photoelectric or chemical effects in ordinary matter. Gamma rays are produced and absorbed by nuclear transitions; their photons have energies high enough to ionize atoms easily and to penetrate dense matter.

The interaction differences are real and physiologically important (gamma rays damage tissue; radio waves do not), but they do not entail differences in propagation speed. The substrate reads all photons identically with respect to propagation; matter reads them differently with respect to absorption.

This is the essential clarification of Section 8, restated here in the context of the spectrum: across the electromagnetic spectrum, frequency varies, propagation speed does not.

10. TIME AS MEASURE, NOT DIMENSION: RECOVERING ARISTOTLE ON FIRM GROUND

The substrate reading invites a reconsideration of one of the most contested concepts in modern physics: time. Modern physics has, in its dominant interpretive register, treated time as a fourth dimension, on a par with the three spatial dimensions, and capable of being "warped" or "bent" by motion or gravity. This treatment has been spectacularly successful in calculation, but it has produced a family of interpretive paradoxes, principally the time-travel paradoxes, that resist resolution within the geometric framework that generates them.

10.1 The Aristotelian Definition

Aristotle, in *Physics* IV.10-14, defined time as "the measure (or number) of motion with respect to before and after." Time, for Aristotle, is not a substance, not a thing, not a place, not a dimension. Time is the count we apply to processes that exhibit before-and-after structure. Where there is motion, there is time as the measure of that motion. Where there is no motion, there is no time, because there is nothing to count.

This definition has often been dismissed as primitive, but careful examination shows it to be precise. Time, in Aristotle's account, is parasitic on change. A clock measures time in the sense that the clock undergoes regular changes (gear movements, oscillations, atomic transitions) that can be counted. The clock is not itself measuring time; the clock is undergoing changes whose count we use to measure other changes.

If all motion stopped, time would stop, not in the sense that an external time-flow would freeze, but in the sense that there would be nothing to count and no measure to apply. Time is the count of motion; no motion, no count, no time.

10.2 Time in Modern Physics

Modern physics has, particularly under the influence of relativity, treated time as a coordinate on a par with spatial coordinates. The combined four-dimensional structure (Minkowski spacetime in special relativity, curved Lorentzian spacetime in general relativity) has been the geometric arena in which physical theories are formulated. Time, in this framework, is something that can be plotted, measured along, and (with motion or gravity) "dilated" or "compressed."

The success of this geometric treatment in calculation is undeniable. Special relativity, general relativity, and quantum field theory all use time as a coordinate, and the empirical predictions are extraordinarily accurate.

The interpretive question is whether time is "really" a dimension, on a par with space, or whether the dimensional treatment is a calculational convenience that does not correspond to time's underlying nature. The physicist concerned only with prediction does not need to settle this question. The philosopher, and the physicist concerned with what the equations are saying, does.

10.3 Time as Measure, in Modern Form

The substrate reading invites a recovery of the Aristotelian definition in modern form. Time, on the substrate reading, is not a dimension. Time is the measure of motion or change, where motion or change is now understood as substrate-perturbation propagation, field-state evolution, or any other process that exhibits a definite temporal ordering.

This is not a rejection of the geometric treatment in calculation. The geometric treatment remains valid as a calculational scheme. What is rejected is the ontological reading that takes time to be a substance or a dimension on a par with space. Time is not a thing; time is what we call the count of changes.

The advantage of this reading is that it dissolves a family of paradoxes that the geometric reading produces.

10.4 Time-Travel Paradoxes

The grandfather paradox, in its standard form, considers a time traveler who returns to the past and prevents his own birth. If he succeeds, he was never born, so he never returned, so the prevention never occurred, so he was born after all. The paradox is irresolvable within the framework that generates it.

On the substrate reading, the grandfather paradox is malformed at its premise. The past is not a place to which one can travel. The past is the name we give to changes that have already occurred and have been registered in the substrate's accumulated state. Once a change has occurred, the substrate's state has evolved; there is no "moment in the past" preserved somewhere as a destination. There is only the substrate as it currently is, with its accumulated history of changes. To "travel to the past" would be to traverse a road to a place that has been consumed in the very process of becoming the present. There is no road and there is no place.

Likewise, the future is not a place. The future is the name we give to changes that have not yet occurred, that exist only as potencies of the substrate. They have no substantial existence; they are dispositions of the present substrate to

evolve in certain ways given certain conditions. To "travel to the future" would be to traverse a road to a place that does not yet exist. There is no place to travel to.

A separate question concerns the closed timelike curves (CTCs) that appear in certain general-relativistic solutions, including Gödel spacetime and traversable wormholes supported by exotic matter. Whether such geometries are physically realizable is genuinely open within general relativity itself. The Hawking chronology protection conjecture argues that quantum effects prevent the formation of CTCs in physical spacetimes, and this conjecture has substantial theoretical support but is not proven. Theorems by Tipler and others establish constraints on CTC formation but leave open whether sufficiently exotic configurations could in principle produce them.

The substrate reading offers a perspective consistent with chronology protection but does not derive it. If the substrate is a physical entity whose evolution is constrained by internal consistency (no state-evolution-paths producing genuine contradictions), then CTCs supporting actual time travel would be ruled out by consistency requirements on the substrate's own dynamics. Whether this consistency requirement amounts to a derivable principle or merely a desideratum awaiting more careful analysis is itself an open question. The honest position is that CTC realizability is empirically and theoretically unsettled, that the substrate reading prefers chronology protection on consistency grounds, and that the question awaits further progress in quantum gravity for definitive resolution. The earlier formulation declaring CTCs "unphysical" was too strong; what should be said is that the substrate reading provides reasons to expect chronology protection without conclusively proving it.

10.5 Time Dilation Without Dimensional Bending

Time dilation, on the substrate reading, is not a dilation of time but a dilation of process rates, as discussed in Section 9.5. A clarification on the comparison is required. To say the traveling twin's processes "proceeded at slower rates" than his stationary brother's might appear to require an external time against which both rates are measured, which would contradict the paper's commitment to time as the measure of motion rather than as a substance.

The clarification is that the comparison is not made in any external time. The comparison is made by the twins themselves at their reunion, by counting internal cycles accumulated between two events both twins share: their parting and their reunion. Each twin's count is a fact about his own worldline. The traveling twin counts fewer cycles; the stationary twin counts more. The difference between these two cycle-counts is the empirical content of time dilation. No external clock, no absolute time, no third-party reference is required. The shared boundary events bound the comparison interval; the internal cycle-counts populate the comparison.

What is "slower," then, is the rate of cycle-accumulation per shared-interval, not the rate of cycle-accumulation per external time. This is consistent with the Aristotelian definition of time as the measure of motion. Each twin counts his own changes; the counts differ along different worldlines between the same pair of shared events; this is exactly the empirical content of relativistic time dilation, framed without recourse to an external time substance.

The traveling twin returns younger because his cycle-count between the shared boundary events is lower than his brother's. He aged less because he counted less internal change between the same shared events. Time, as such, did not warp for him; what changed was the rate at which his localized processes registered changes during the traveling phase. The dilation is real and observable; it is a path-dependent rate of process-accumulation between shared boundary events, not a dilation of any external time substance. There is nothing mysterious here once the comparison is properly framed.

11. CURRENT GAPS AND SPECIFIC FALSIFIABLE PREDICTIONS

The substrate-perturbation reading is empirically equivalent to standard physics in essentially every confirmed observation. Its primary contribution is interpretive. There are, however, specific places where the framework's commitments produce falsifiable predictions, and there are specific theoretical gaps within the framework itself that require further development. This section enumerates both, with attention to the precision required for the predictions to be genuinely testable.

11.1 Theoretical Gaps Within the Framework

The framework is not a complete theory of physics. Several internal questions remain open and require further development.

Derivation of Bell correlations from substrate dynamics. The framework restates the entanglement structure as a feature of the field's joint quantum state, consistent with quantum field theory. It does not derive the specific quantitative violations of Bell inequalities (the cosine-squared correlation form, the precise CHSH-bound saturation pattern) from substrate first principles in a way that goes beyond standard QFT calculation. Whether the substrate ontology can produce a deeper derivation of the Bell correlations is an open theoretical question.

Topological characterization of massive fermions. The framework treats massive fermions as topologically stable knot configurations of the underlying substrate fields. Faddeev-Niemi-style Hopf-soliton models offer one route to mathematizing this, with the lepton mass hierarchy potentially mapping to the energy hierarchy of distinct knot topologies. The full mathematical characterization (which knots correspond to which leptons, what role color and isospin play in the topological structure, how generations arise from topological complexity) remains incomplete.

Quantum-gravitational unification. The framework treats the substrate as both a quantum entity (whose excitations are field quanta) and a geometric entity (whose configuration produces gravity), but the explicit unification of these aspects awaits further development. The framework is compatible with multiple quantum-gravity programs (string theory, loop quantum gravity, asymptotic safety) and does not currently select among them on substrate-ontological grounds.

Functional form of dark energy $w(z)$. The framework predicts $w \geq -1$ strictly (Section 11.3 below), but the specific functional form of $w(z)$ within this constraint requires more detailed substrate-tension modeling.

Substrate behavior under Planck-scale conditions. Behavior at energy densities approaching the Planck scale, at the cosmological singularity, or in regions of extreme curvature is not yet fully characterized within the framework. The substrate ontology suggests such regimes should produce specific signatures, but the calculations have not been carried out in detail.

11.2 MOND Acceleration Scale

The framework predicts a specific value for the MOND-like acceleration scale a_0 , derived from substrate-Hubble coevolution rather than fitted to data.

Specific prediction. $a_0 = (2/3) \cdot (1/\sqrt{3}) \cdot \sqrt{(\Omega_{DE}/3)} \cdot c \cdot H_0 \approx 0.183 \cdot c \cdot H_0$

For $H_0 = 67.4$ km/s/Mpc (Planck CMB-derived value): $a_0 \approx 1.13 \times 10^{-10}$ m/s². For $H_0 = 73$ km/s/Mpc (SH0ES local-distance-ladder value): $a_0 \approx 1.22 \times 10^{-10}$ m/s².

Empirical anchor. The SPARC database analysis (McGaugh, Lelli, and Schombert 2016) yields $a_0 \approx 1.2 \times 10^{-10}$ m/s² across 175 galaxies, agreeing with the framework's prediction to within current observational precision.

Test. Continued precision measurement of galaxy rotation curves (SPARC and successors), wide-binary star kinematics (Chae 2023 reports 4σ MOND-like behavior in low-acceleration binaries), and galaxy cluster dynamics. Future surveys (Euclid, Roman Space Telescope) will tighten constraints.

Falsification. Robust measurement of a_0 deviating from the predicted range by more than 20% across multiple independent methods would falsify the framework's identification of dark-sector physics as substrate-tensional structure.

Timeframe. Active testing 2024-2030.

11.3 Phantom Dark Energy Forbidden

The framework predicts that dark energy's equation-of-state parameter $w(z)$ is bounded from below by -1 strictly: phantom dark energy is forbidden by the framework's substrate-thermodynamic consistency requirements.

Specific prediction. $w(z) \geq -1$ for all redshifts z accessible to observation. Quintessence-like behavior ($-1 < w \leq 0$) is permitted; phantom behavior ($w < -1$) is not, at any redshift.

Current empirical status. This prediction is in active empirical tension with recent observational results and warrants honest engagement rather than deferred testing. DESI DR1 (Adame et al. 2024) showed a preference for dynamical dark energy over Λ CDM at moderate significance using the Chevallier-Polarski-Linder (CPL) parametrization, with the best-fit $w_0 w_a$ CDM model exhibiting phantom-barrier crossing in the recent past (transition from w

$w < -1$ in the past to $w > -1$ today). DESI DR2 (Adame et al. 2025) strengthened this preference, with combined DESI+Planck/ACT/SPT+SN (PantheonPlus, Union3, or DESY5) datasets all robustly favoring the Quintom-B regime (past phantom transitioning to present quintessence) at significances reported in the range 2.5σ - 3.9σ depending on dataset combination.

The framework's strict $w(z) \geq -1$ prediction is therefore at risk now, not in 2027-2030 as the earlier draft of this paper implied. Three caveats apply, none of which softens the situation to comfort. First, the present-day CPL central values for w_0 have shifted substantially away from -1 , not toward it. DESI DR2 + CMB combined with the three SN compilations gives $w_0 \approx -0.83 \pm 0.06$ (Pantheon+), $w_0 \approx -0.65 \pm 0.10$ (Union3), and $w_0 \approx -0.73 \pm 0.06$ (DESY5) within the w_0w_a CDM parametrization. The constant- w pivot value w_{pivot} remains consistent with -1 to within roughly ± 0.03 (Cortês and Liddle 2025) when integrated over the observational window, but this is a feature of where the pivot redshift falls rather than evidence that w_0 itself is consistent with -1 . The detection significance lives in the w_a parameter, with the joint w_0 - w_a contour displaced from the cosmological-constant point at significances of 2.8σ (Pantheon+), 3.8σ (Union3), and 4.2σ (DESY5), with combined three-sample exclusion of approximately 3.1σ . Second, the CPL parametrization has been criticized for its prior choices and parameter degeneracies, and quintom models without strict phantom crossing remain compatible with the data at the 2σ level (Gialamas et al. 2025). Third, model-independent reconstructions sensitive to $z < 0.1$ supernovae have raised the question whether the apparent dynamical signal reflects late-time cosmological evolution or unmodeled local-universe systematics.

A counter-current development worth noting cuts in the framework's favor without resolving the broader question. Möller et al. 2025 (arXiv:2511.07517) introduce the DES-Dovekie recalibration of the DES Y5 supernova sample, incorporating improved photometric cross-calibration, white-dwarf cross-calibration between DES and low-redshift surveys, retraining of the SALT3 light-curve model, and correction of a numerical approximation in the host galaxy color law. Combining DES-Dovekie with Planck/ACT/SPT CMB data and DESI DR2, the w_0w_a CDM fit yields $w_0 = -0.803 \pm 0.054$ and $w_a = -0.72 \pm 0.21$, with the Λ CDM exclusion significance reduced from the original 4.2σ for DES-SN5YR to 3.2σ for DES-Dovekie. The Bayesian model preference is approximately 5:1 in favor of w_0w_a CDM, which the authors describe as a weak rather than strong preference. If this recalibration holds up under independent scrutiny, the DESY5-driven component of the dynamical-dark-energy preference weakens substantially, and the framework's prediction faces less acute pressure than the unrecalibrated 4.2σ figure suggested. This does not eliminate the tension; the Pantheon+ and Union3 contributions remain at 2.8σ and 3.8σ respectively. It does indicate that the headline 4.2σ DESY5 number was driven in part by calibration systematics now believed to be corrected, and that the empirical

situation is genuinely in flux as supernova systematics continue to be scrutinized.

Falsification. The framework's strict $w(z) \geq -1$ prediction is falsified if independent and robust detection of $w < -1$ at any redshift, persistent across parametrization choices and supernova compilations, reaches 5σ significance. The current $2.5\text{--}3.9\sigma$ tension does not yet falsify the prediction but does constitute substantial empirical pressure on it. If DESI DR3 and Euclid+Roman+LSST combined data sustain or strengthen the phantom-crossing preference at 5σ across multiple independent SN compilations, the framework's substrate-thermodynamic identification of dark energy will require either substantial revision or abandonment.

Test. DESI DR3 (expected 2027), Euclid full data (2027-2030), Roman Space Telescope dark energy survey (2027 launch), and LSST/Vera Rubin Observatory (operational 2026) will jointly determine whether the phantom-crossing preference strengthens or weakens. ACT and Simons Observatory CMB measurements will refine the early-universe anchor. If the w_a evolution preference disappears with improved data quality, the framework's prediction survives intact; if it strengthens, the framework requires substantial reconsideration.

Timeframe. Active empirical pressure now; definitive answer 2027-2030 with Stage-IV dark energy surveys.

11.4 MOND Interpolation Function

The framework predicts a specific functional form for the MOND interpolation between Newtonian and modified-gravity regimes, derived from Fourier-duality of substrate gradient interference.

Specific prediction. The interpolation function is $\mu(x) = x / \sqrt{1 + x^2}$, where $x = g_N / a_0$ is the ratio of the Newtonian gravitational acceleration to the framework's predicted a_0 scale.

This form is empirically distinguishable from the alternative simple-form interpolation $\mu(x) = x / (1 + x)$, which is favored in some classical MOND treatments.

Empirical anchor. SPARC galaxy rotation curve analyses give the $\sqrt{}$ -form a modest preference over the simple-form in some treatments, though this preference is dataset-dependent and parametrization-dependent rather than a clean across-the-board result. Lelli, McGaugh, and Schombert analyses across multiple functional families have shown competing preferences, with the simple-form $\mu(x) = x/(1+x)$ often performing comparably or in some cases preferentially across subsamples. The framework's commitment to the $\sqrt{}$ -form is therefore a specific testable claim rather than a confirmed empirical regularity, and the

framework will stand or fall on whether higher-precision rotation-curve data (extended SPARC samples, gas-rich dwarfs, ultra-low surface brightness galaxies) and wide-binary kinematics in the deep-MOND regime confirm or refute the $\sqrt{\cdot}$ -form preference. Tully-Fisher analyses give consistent fits using $v_{\infty}^4 = G \cdot M_b \cdot a_0$ in both functional forms; the discrimination between $\sqrt{\cdot}$ -form and simple-form requires precision rotation-curve data at intermediate accelerations near the a_0 scale.

Test. Continued SPARC analysis with additional galaxies, gas-rich dwarf galaxy surveys, and ultra-low surface brightness galaxy observations. Wide-binary star kinematics at very low accelerations.

Falsification. A statistically significant fit-quality preference for an alternative interpolation function over the $\sqrt{\cdot}$ -form across multiple independent samples would falsify this prediction.

Timeframe. Active testing ongoing; maturity expected 2025-2030.

11.5 Hubble Tension as Environmental Effect

The framework predicts that the persistent Hubble tension (the discrepancy between Planck-CMB-derived $H_0 \approx 67.4$ km/s/Mpc and local-distance-ladder $H_0 \approx 73$ km/s/Mpc) may reflect a real physical effect of substrate-tensional structure correlating with local matter density, in addition to or instead of measurement artifacts.

Specific prediction. Local H_0 measurements should vary systematically with the local matter density. The framework predicts a fractional variation of order $\Delta H_0/H_0 \sim 0.05 \cdot \delta$, where δ is the local matter density contrast ($\delta = (\rho - \bar{\rho})/\bar{\rho}$). The sign of this prediction is that overdense regions show elevated local H_0 and underdense regions show depressed local H_0 .

Empirical anchor and complication. The current SH0ES vs Planck tension is approximately 8% at 5σ . The framework's environmental prediction can account for a substantial fraction of this only if the SH0ES sampling region is overdense by $\delta \sim 1$, which is a strong claim about the local universe's matter distribution. The observational status is genuinely contested. Some studies (for instance Keenan, Barger, and Cowie 2013; Wong et al. 2022 on Hubble flow corrections) suggest the local volume out to roughly 300 Mpc may be underdense (the "KBC void" hypothesis), in which case the framework's prediction would have the opposite sign from what is needed to account for the SH0ES result. Other studies (Cosmicflows-4 analyses, peculiar-velocity reconstructions) suggest mild local overdensity at smaller scales but density profiles depending sensitively on smoothing scales and tracer choices. The framework's ability to account for the observed tension therefore depends on observational determinations of local density structure that are not yet definitively settled.

The honest position is that the framework predicts an environmental dependence with a specific sign (denser $\rightarrow$ higher local H_0) and rough magnitude ($\Delta H_0/H_0 \sim 0.05 \cdot \delta$). Whether this prediction can account for the observed SH0ES vs Planck tension depends on the actual density structure of the local volume, which is itself an active observational question. If the local volume is found to be underdense and SH0ES still measures elevated local H_0 , the framework's environmental account fails, and the Hubble tension would remain unexplained on substrate-environmental grounds.

Test. Distance ladder measurements in cosmic voids (Type Ia supernovae in low-density environments), comparison of H_0 measured in galaxy-cluster vs field environments, time-delay cosmography in different cosmic environments (H0LiCOW, TDCOSMO), and improved characterization of local-volume density profiles (Cosmicflows-5 and successors). Any environmental dependence at the predicted magnitude with the predicted sign would support the framework; absence of such dependence would falsify the environmental component of the framework's account.

Falsification. Demonstration that local H_0 is independent of local density at precision better than 1% across measurement environments differing in δ by order unity, or that the dependence has opposite sign from the prediction (denser $\rightarrow$ lower H_0), would falsify the framework's environmental account.

Timeframe. Likely partial resolution 2025-2028 with combined Stage-IV surveys; full settlement of local-volume density profiles may take longer.

11.6 Speed of Gravity vs Speed of Light Differential

The framework predicts that gravitational waves and electromagnetic waves propagate at exactly the same speed in vacuum, with no detectable difference under any conditions in normal substrate regimes.

Specific prediction. $|c_{\text{gw}} - c_{\text{em}}| / c < 10^{-16}$ in any future binary merger event with electromagnetic counterpart at cosmological distances.

Empirical anchor. GW170817 binary neutron star merger constrained $|c_{\text{gw}} - c_{\text{em}}| / c < 7 \times 10^{-16}$ from the 1.7-second arrival-time differential over 130 million light-years of travel. This is consistent with zero deviation.

Test. Future binary neutron star mergers detected by LIGO-Virgo-KAGRA-LIGO India and successor observatories (Einstein Telescope, Cosmic Explorer), particularly events at higher redshifts that extend the baseline. Coincident gamma-ray-burst observations from Fermi GBM, INTEGRAL, and successors.

Falsification. Robust measurement of $c_{\text{gw}} \neq c_{\text{em}}$ at any precision better than 10^{-15} would falsify the framework's prediction of universal massless-excitation propagation at c .

Timeframe. Ongoing; multiple events per year expected with next-generation observatories starting late 2020s.

11.7 Neutrino Mass Ordering and Sum

The framework predicts normal mass ordering for the neutrinos, with the lightest mass close to zero and the sum near the minimum allowed by oscillation constraints.

Specific prediction. Normal ordering: $m_1 < m_2 < m_3$ with $m_1 < 10$ meV. Sum of masses $\Sigma m_\nu \approx 58\text{--}70$ meV, close to the minimum allowed by Δm^2_{atm} and Δm^2_{sol} oscillation parameters.

Empirical anchor. Current oscillation data establish the mass-squared splittings but leave the absolute scale and ordering open. Cosmological constraints from Planck + BAO + LSS give $\Sigma m_\nu < 120$ meV at 95% confidence. The framework's prediction occupies the lower end of the allowed range.

Test. KATRIN tritium endpoint measurement (direct kinematic mass), reaching sensitivity to $m_\beta \sim 200$ meV. JUNO medium-baseline reactor experiment (mass ordering at $3\text{--}4\sigma$ within 6 years of operation, started 2024). DUNE long-baseline experiment (mass ordering at $>5\sigma$ by mid-2030s). Hyper-Kamiokande (operational 2027). Cosmological constraints from Euclid, LSST, Simons Observatory, and CMB-S4 should reach Σm_ν sensitivity ~ 30 meV.

Falsification. Robust measurement of inverted ordering at $>3\sigma$ from JUNO or DUNE, or measurement of $\Sigma m_\nu > 100$ meV from cosmology, would falsify the framework's prediction.

Timeframe. Mass ordering: 2026-2030 (JUNO) and 2030-2035 (DUNE/Hyper-K). Sum constraints: 2027-2032 (Stage-IV cosmology).

11.8 Lepton CP-Violating Phase

The framework predicts a specific value for the Dirac CP-violating phase in the lepton sector, derived from topological-charge constraints on the three-generation lepton structure.

Specific prediction. $\delta_{\text{CP}} \approx -\pi/2$ (equivalently, 270° or $3\pi/2$ modulo 2π), with the negative branch preferred over positive (so excluding δ_{CP} near $+\pi/2$).

Empirical anchor. T2K 2020-2024 analyses prefer δ_{CP} near $-\pi/2$ at approximately $2\text{--}3\sigma$. NOvA results have been more ambiguous but consistent with non-zero CP violation.

Test. DUNE long-baseline neutrino experiment (sensitivity to δ_{CP} at $>5\sigma$ for substantial CP violation, expected 2030-2035). Hyper-Kamiokande long-baseline

measurements (operational 2027, similar precision goals). Combined T2K + NOvA + reactor anti-neutrino constraints.

Falsification. Robust measurement of $\delta_{\text{CP}} > 0$ at $>3\sigma$, or measurement of δ_{CP} consistent with zero or π at $>3\sigma$, would falsify the framework's prediction of $-\pi/2$ region.

Timeframe. Definitive measurement 2030-2035.

11.9 Fine-Structure Constant Variation

The framework allows for, but does not strictly require, small temporal variation of the fine-structure constant α as substrate-tension structure evolves with cosmic expansion.

Specific prediction. $|\Delta\alpha/\alpha| < 10^{-6}$ at $z \sim 2$ in any direction. The framework is consistent with no detectable variation at current precision but does not forbid small variation at the 10^{-7} level.

Empirical anchor. Webb et al. 2011 reported a possible spatial variation of α at the 10^{-5} level, but the result has been contested by subsequent analyses (Murphy et al. 2022 finds no variation at 10^{-5} precision in higher-quality spectra). Current best constraints from quasar absorption spectroscopy give $|\Delta\alpha/\alpha| < \text{a few} \times 10^{-6}$ at $z \sim 2-4$.

Test. ESPRESSO spectrograph at the Very Large Telescope (precision 10^{-7} achievable), atomic clock comparisons over multi-year timescales (Hg+/Al+ comparisons reaching 10^{-18} per year drift sensitivity), 21 cm cosmology probes of α at $z \sim 6-30$.

Falsification. Robust detection of $|\Delta\alpha/\alpha| > 10^{-5}$ would require substantial revision of the framework's substrate-coupling-constancy assumption. Robust detection at the 10^{-7} level would constrain substrate-tension evolution models.

Timeframe. ESPRESSO results 2025-2030; atomic clock cumulative constraints ongoing.

11.10 Lorentz Invariance Across the Electromagnetic Spectrum

The framework predicts that all massless field excitations propagate at exactly c regardless of frequency, energy, or polarization, as a consequence of the substrate's Lorentz-invariance under arbitrary electromagnetic excitations.

Specific prediction. $\Delta c/c < 10^{-15}$ across the entire electromagnetic spectrum from radio (10^{-9} eV) to ultra-high-energy cosmic-ray photons (10^{20} eV). The framework predicts no frequency-dependent dispersion in vacuum.

Empirical anchor. GW170817 multi-messenger observations confirmed $\Delta c/c < 10^{-15}$ across the spectrum from gamma-ray to optical to radio. Fermi gamma-ray observations of GRB 090510 constrained Lorentz invariance to similar precision at very high photon energies.

Test. Continued multi-messenger astronomy, particularly observations of high-energy astrophysical sources at cosmological distances. Cherenkov Telescope Array (CTA) observations of distant blazars and gamma-ray bursts. IceCube astrophysical neutrino observations cross-correlated with electromagnetic counterparts.

Falsification. Robust detection of frequency-dependent dispersion of light in vacuum, at any precision better than 10^{-15} , would falsify the framework's substrate-uniformity prediction.

Timeframe. Ongoing; cumulative precision improving with each multi-messenger event.

11.11 Force Hierarchy as Phase-Transition Squared Ratio

The framework predicts a specific structural origin for the 38-order-of-magnitude hierarchy between the strong nuclear force and gravitational force at the proton scale.

Specific prediction. The dimensionless ratio characterizing the strong-gravitational hierarchy at the proton scale exhibits the squared-ratio structure $\alpha_s \cdot (M_P / m_p)^2$, where M_P is the Planck mass, m_p is the proton mass, and α_s is the strong coupling evaluated at the relevant QCD scale. The two squared phase-transition scales (M_P from gravitational substrate formation at the Planck epoch, m_p set by QCD confinement at temperatures around 155 MeV) are framework-predicted to be independent, and the dimensionless ratio of their squares dominates the hierarchy.

Numerical relation. In natural units, $G_N \cdot m_p^2 = (m_p / M_P)^2$, so $\alpha_s / (G_N \cdot m_p^2) = \alpha_s \cdot (M_P / m_p)^2$. With $(M_P / m_p)^2 \approx 1.69 \times 10^{38}$ and $\alpha_s \approx 0.118$ evaluated at the Z scale, the resulting product is approximately 2.0×10^{37} . With α_s evaluated near the QCD confinement scale (where the running coupling becomes order unity in conventional perturbative treatments, with formal divergence at Λ_{QCD}), the product approaches the empirical ratio 1.7×10^{38} . The precise numerical value is therefore scale-dependent in a way the earlier draft of this paper did not specify clearly. The prediction's structural content is the squared-ratio $(M_P/m_p)^2$ architecture; the exact factor of α_s is a scale-dependent dressing rather than an additional predictive coincidence.

Empirical anchor. The empirical ratio $\alpha_s / (G_N \cdot m_p^2)$ is approximately 1.7×10^{38} when α_s is evaluated near the confinement scale. The order-of-magnitude agreement with $(M_P/m_p)^2$ is robust; the precise multiplicative factor of α_s

depends on the renormalization scale and on the prescription used to handle non-perturbative regimes. The framework's claim is that the squared-ratio architecture dominates the hierarchy structurally, not that $\alpha_s \cdot (M_P/m_p)^2$ agrees numerically to within a few percent at any specific scale choice.

Test. Continued precision measurement of fundamental constants (G_N from torsion-balance and atom-interferometry experiments, m_p from atomic-mass spectrometry, α_s from lattice QCD and high-precision collider phenomenology). Faddeev-Niemi-style numerical lattice computation of QCD-scale soliton energies could allow direct calculation of m_p from substrate-topological inputs rather than fit, providing a more rigorous test of the framework's claim that m_p is set by a confinement-scale phase-transition independent of the gravitational sector.

Falsification. A robust theoretical or empirical demonstration that the proton mass cannot be derived from a confinement-scale phase-transition independent of the gravitational sector, or that the squared-ratio $(M_P/m_p)^2$ architecture fails to dominate the hierarchy at any scale, would weaken this identification. Detection of additional dimensionless factors of order 10^4 or larger entering the hierarchy in ways not accountable by the framework's structural argument would constitute falsification.

Timeframe. Ongoing; lattice QCD computation of soliton energies is the most likely route to direct test (2025-2035 timeframe).

11.12 Summary Table of Predictions

The following table consolidates the framework's specific falsifiable predictions, the experiments or observations that test them, the precision required for falsification, and the expected timeframes for definitive results.

#	Prediction	Specific Value	Test/ Experiment	Falsification Threshold	Timeframe
1	MOND acceleration scale a_0	$0.183 \cdot c \cdot H_0 \approx 1.13\text{--}1.22 \times 10^{-10} \text{ m/s}^2$	SPARC RAR analysis, wide-binary kinematics, galaxy cluster dynamics	Deviation > 20% across independent methods	2024–2030
2	Phantom dark energy forbidden	$w(z) \geq -1$ for all z	DESI DR2/ DR3, Euclid, Roman, LSST + SN compilations	Phantom-crossing at > 5σ across parametrizations and SN datasets	Active tension now ($2.5\text{--}3.9\sigma$); definitive 2027–2030
3	MOND interpolation function	$\mu(x) = x / \sqrt{1+x^2}$; modestly preferred in some SPARC analyses, contested in others	Extended SPARC, gas-rich dwarfs, ultra-LSB galaxies, wide-binary kinematics	Alternative form fits significantly better across multiple independent samples	Ongoing through 2030
4	Hubble tension environmental	$\Delta H_0/H_0 \sim 0.05 \cdot \delta$; sign: denser $\rightarrow$ higher H_0	Distance ladders in voids vs clusters, time-delay cosmography, local-volume density profiling	No environmental dependence at predicted level, OR opposite sign found	2025–2028 (partial); local density itself contested
5	Speed of gravity = speed of light	$\Delta c/c$ (gw vs em) < 10^{-16}	Future BNS mergers with EM counterparts	Deviation > 10^{-15} at any precision	Ongoing
6	Neutrino mass ordering	Normal ordering, $m_1 < 10 \text{ meV}$	KATRIN, JUNO, DUNE, Hyper-K	Inverted ordering at > 3σ	2026–2035
7	Neutrino mass sum	$\Sigma m_\nu \approx 58\text{--}70 \text{ meV}$	Cosmology (Euclid,		2027–2032

#	Prediction	Specific Value	Test/ Experiment	Falsification Threshold	Timeframe
			LSST, SO, CMB-S4)	$\Sigma m_\nu > 100$ meV from cosmology	
8	CP-violating phase δ_{CP}	$\approx -\pi/2$ (270°, negative branch)	DUNE, Hyper-K, T2K continuation	$\delta_{CP} > 0$ at $> 3\sigma$	2030-2035
9	Fine-structure constant α	$\Delta\alpha/\alpha < 10^{-6}$ in magnitude at $z \sim 2$	ESPRESSO, atomic clocks, 21 cm cosmology	Detection $> 10^{-5}$ at robust significance	2025-2030
10	Lorentz invariance of c	$\Delta c/c < 10^{-15}$ across full EM spectrum	Multi-messenger astronomy, CTA, IceCube	Frequency-dependent dispersion at any precision	Ongoing
11	Force hierarchy structure	$\alpha_s \cdot (M_P / m_p)^2$ with squared-ratio architecture; $(M_P/m_p)^2 \approx 1.69 \times 10^{38}$ as structural core	Precision constants, lattice QCD soliton energies	Squared-ratio architecture fails; m_p not derivable from confinement-scale phase transition	2025-2035

The eleven predictions above span scales from sub-atomic to cosmological. Several are testable with currently operating experiments; others await next-generation observatories. They are not all equally distinguishing of the framework relative to standard physics. Predictions 5 (speed of gravity equals speed of light to extreme precision), 9 (fine-structure constant variation below 10^{-6} at $z \sim 2$), and 10 (Lorentz invariance of c across the spectrum) are essentially what standard physics would also predict; the framework's commitment to these predictions is consistent with the standard reading rather than uniquely entailed by substrate ontology. They serve as consistency tests rather than as distinguishing experiments. Predictions 1, 2, 3, 4, 6, 7, 8, and 11 carry substrate-specific content that would not be entailed by standard physics treatments lacking the framework's particular commitments.

The framework is genuinely falsifiable on the distinguishing predictions. Robust positive results on any of the falsification thresholds for predictions 1-4, 6-8, or

11 would require either substantial revision or abandonment of the framework's specific commitments. Prediction 2 (phantom dark energy forbidden) is in active empirical tension at the $2.5\text{-}3.9\sigma$ level with current DESI-era observations and may force the issue earlier than the formal Stage-IV timeline. Prediction 11 (force hierarchy as squared phase-transition ratio) is structural rather than precisely numerological, with the $(M_P/m_p)^2$ architecture being the framework's predictive content and the precise α_s factor depending on scale conventions.

Honest empirical engagement requires acknowledging that the framework is not uniformly distinguished from standard physics across all eleven predictions, and that the most distinguishing predictions (1-4 and 11 in particular) carry the most substantial empirical risk. The next decade of observation should settle the framework's empirical status on these distinguishing predictions, and the framework will stand or fall on those results rather than on the consistency-test predictions that any reasonable physics would equally predict.

12. CONCLUSION: A CONTINUOUS-MEDIUM READING WITH SPECIFIC EMPIRICAL STAKES

Twenty-five centuries of optical inquiry, from the visual rays of Empedocles to the field excitations of quantum field theory, exhibit a partial but real convergence on a structural intuition: light is a perturbation propagating through a continuous medium that is space itself. The convergence is partial because the specific theories embedding this intuition have differed substantially. Aristotle's instantaneous activation, Huygens's mechanical ether, Maxwell's electromagnetic wave, and the photon as field-quantum are not the same theory at different levels of formal precision. They differ on essential points: instantaneous vs finite- c propagation, mechanical vs Lorentz-invariant medium, classical wave vs quantum excitation. What unifies them is the commitment to a continuous medium and the rejection of the void as the locus of optical phenomena. That is a real thread, but it is a thread, not a single embroidered theory.

The interpretive position the present paper develops is the neo-Lorentzian reading of relativistic and quantum field-theoretic physics. The substrate is taken to be physically real and continuous. Lorentz-invariance of the field equations is treated as an emergent feature of the substrate's dynamics rather than as a metaphysical primitive. Empirical predictions are identical to those of standard special relativity in tested regimes; ontological commitments are stronger and clearer. The substrate is not a substance among substances; it is space, considered as the entity that admits perturbations and that has its own bodily extension distinct from the propagation of those perturbations within it.

The reading carries specific consequences. Wave-particle duality is reframed as the meeting of a continuous substrate with discrete matter at the point of registration, with no further interpretive paradox required. Time dilation is reframed as a path-dependent rate of internal-process accumulation between shared boundary events, without invoking time as a substance or dimension. Cosmological metric expansion is understood as the substrate's bodily extension, distinct from and not bounded by the propagation rate of perturbations within it. Black-hole event horizons are global features of the substrate's geometric configuration rather than local field-strength extrema. Entanglement is restated in geometric terms (one substrate, one non-separable field-state) but not derived in a way that goes beyond standard quantum field theory. Time-travel paradoxes dissolve at their premise (the past is not a place; the future is not a place), with closed timelike curves marked as theoretically open and consistency-protected on substrate grounds.

The framework is empirically equivalent to standard physics in essentially all currently tested regimes. Its primary contribution is interpretive. But it is not purely interpretive. Section 11 enumerated eleven specific empirical commitments. Eight of these (predictions 1, 2, 3, 4, 6, 7, 8, and 11) carry substrate-specific content that would not be entailed by standard physics treatments lacking the framework's particular commitments. Three (predictions 5, 9, and 10) are essentially restatements of standard expectations that serve as consistency tests rather than as distinguishing experiments. Among the distinguishing predictions, prediction 2 (phantom dark energy forbidden) is in active empirical tension with current DESI-era observations at the $2.5\text{-}3.9\sigma$ level, with the framework's strict $w(z) \geq -1$ prediction at risk of falsification within the next several years rather than at the formal Stage-IV timeline.

The honest position is that the substrate-perturbation reading articulates a genuine ontology that is empirically underdetermined relative to standard interpretations in the regime of confirmed observations and that becomes empirically distinguishable in the regime of the distinguishing predictions enumerated above. If those predictions are confirmed, the substrate-perturbation reading gains substantial empirical traction. If they are falsified, the framework requires substantial revision or abandonment of the relevant commitments. The framework is not protected from empirical refutation by its interpretive character; the interpretive commitments produce predictive consequences in regimes where current physics is not definitive, and those consequences are testable.

The phantom dark energy question deserves particular attention. The framework's commitment to $w(z) \geq -1$ is not a marginal claim that can be quietly retracted. It follows from substrate-thermodynamic consistency requirements that are central to the framework's substrate ontology. If DESI DR3 and Stage-IV surveys sustain the phantom-crossing preference at 5σ across multiple parametrizations and supernova compilations, the framework's identification of dark energy as substrate-tensional structure fails, and a substantial portion of the framework's cosmological apparatus requires reconsideration. This is not a minor adjustment; it is a falsification at the level of one of the framework's central empirical claims. The framework should not pretend otherwise.

What this paper has tried to provide is an honest articulation of the geometric content that runs as a partial thread through twenty-five centuries of optical inquiry, the specific interpretive commitments that the present neo-Lorentzian reading involves, the dissolution of several interpretive paradoxes that the standard reading leaves open, and a specific set of empirical commitments that distinguish the framework from purely interpretive alternatives. The reading is older than its current articulation; the predictions are new and specific; the framework stands or falls on whether observation confirms or refutes its distinguishing claims over the next decade.

The substrate is real and continuous; the photon is its excitation; the propagation is at c ; the substrate's own dynamics produce the cosmological expansion that exceeds c . Time is the measure of process accumulation along worldlines, not a dimension. The medium is space, considered with respect to its electromagnetic properties. These commitments produce specific testable predictions, several of which are already in tension with current observations and several of which await Stage-IV-class data to settle. The framework awaits empirical adjudication on those predictions; the interpretive scaffolding is ready for the test, and the test will not be deferred indefinitely.

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Manuscript prepared for external publication. Self-contained treatment of the nature of light, intended for educated readers across the natural sciences and philosophy of science. Revised version, May 2026, addressing first-round audit-driven structural concerns, second-round audit-driven empirical and mathematical corrections (DESI evidence engagement, force-hierarchy α_s scale specification, substrate-rest-frame identification, Hubble-tension density-status acknowledgment), third-round audit-driven precision corrections (DESI DR2 CPL central-value reporting, DES-Dovekie recalibration acknowledgment, SPARC interpolation-function honesty), and adding Section 11 on falsifiable predictions including a summary table of testable claims. Internal framework citations are limited to a small number of parenthetical cross-references, retained for the convenience of readers familiar with the broader research program from which this paper emerges.